Applied Mathematics 205

Unit 2. Numerical Linear Algebra

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Motivation

- Scientific Computing relies on Numerical Linear Algebra
- We often reformulate problems as $Ax = b$
- Examples from Unit 1:
  - interpolation (Vandermonde matrix) and linear least squares (normal equations) are naturally expressed as linear systems
  - Gauss–Newton method involves approximating a nonlinear problem with a sequence of linear systems
- We will see more applications of linear systems (Numerical Calculus, Optimization, Eigenvalue problems)

Motivation

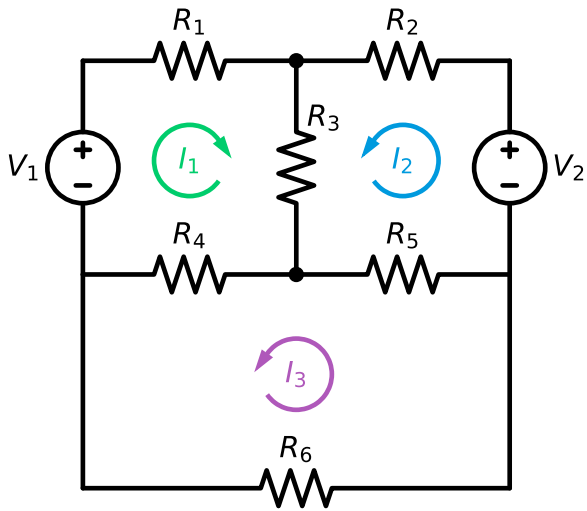
- The goal of this Unit is to cover:
  - concepts from linear algebra relevant for Scientific Computing
  - stable and efficient algorithms for solving $Ax = b$
  - algorithms for computing factorizations of A
    that are useful in many practical contexts (LU, QR)
- First, we discuss some practical cases where $Ax = b$
  arises directly in mathematical modeling of physical systems

Example: Electric Circuits

- Linear systems describe circuits consisting of voltage sources and resistors
  - Ohm's law: Voltage drop V due to current I through a resistor R is

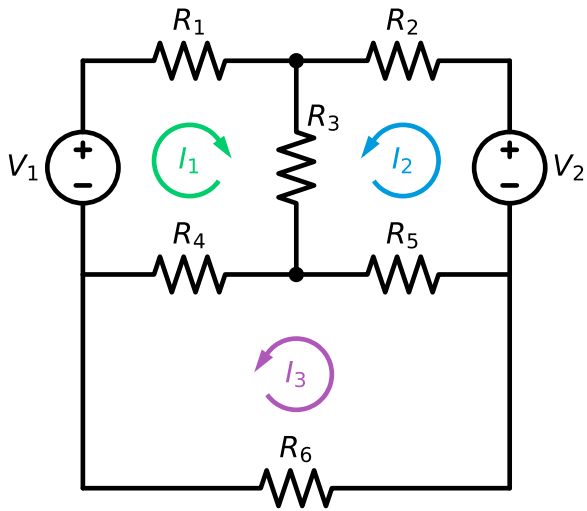
$$V = IR$$

  - Kirchoff's law: Directed sum of the voltages around any closed loop is zero



Example: Electric Circuits

- The circuit has three loops
  - Loop 1
    $R_1 I_1 + R_3(I_1 + I_2) + R_4(I_1 + I_3) = V_1$
  - Loop 2
    $R_2 I_2 + R_3(I_1 + I_2) + R_5(I_2 - I_3) = V_2$
  - Loop 3
    $R_5(I_3 - I_2) + R_4(I_3 + I_1) + R_6 I_3 = 0$



Example: Electric Circuits

- We obtain a linear system for unknown currents I_1, I_2, I_3

$$\begin{bmatrix} R_1 + R_3 + R_4 & R_3 & R_4 \\ R_3 & R_2 + R_3 + R_5 & -R_5 \\ R_4 & -R_5 & R_4 + R_5 + R_6 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \\ I_3 \end{bmatrix} = \begin{bmatrix} V_1 \\ V_2 \\ 0 \end{bmatrix}$$

- Note that the matrix is
  - symmetric, i.e. $a_{ij} = a_{ji}$
  - strictly diagonally dominant, i.e. $|a_{ii}| > \sum_{j \neq i} |a_{ij}|$ (assuming $R_k > 0$)
- Circuit simulators solve large linear systems of this type

Example: Electric Circuits

- Another linear system corresponds to unknown resistances R_i, $i = 1, \ldots, 6$

$$\begin{bmatrix} I_1 & 0 & I_1 + I_2 & I_1 & 0 & 0 \\ 0 & I_2 & I_2 & 0 & -I_3 & 0 \\ 0 & 0 & 0 & I_1 + I_3 & -I_2 & I_3 \end{bmatrix} \begin{bmatrix} R_1 \\ R_2 \\ R_3 \\ R_4 \\ R_5 \\ R_6 \end{bmatrix} = \begin{bmatrix} V_1 \\ V_2 \\ 0 \end{bmatrix}$$

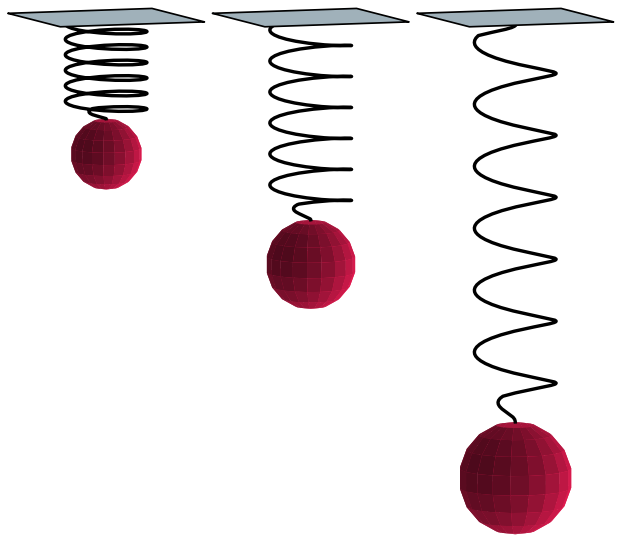
- Note that the matrix has full rank (assuming $I_k \neq 0$)
- The system is underdetermined: 3 equations for 6 unknowns

Example: Structural Analysis

- Common in structural analysis is to use a linear relationship between force and displacement, Hooke's law
- Simplest case is the Hookean spring law

$$F = kx$$

  - k: spring constant (stiffness)
  - F: applied load
  - x: spring extension (displacement)



Example: Structural Analysis

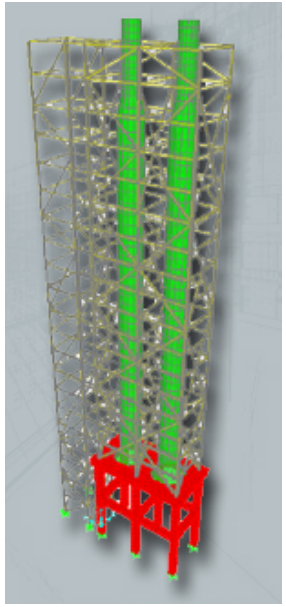
- This relationship can be generalized to structural systems in 2D and 3D, which yields a linear system of the form

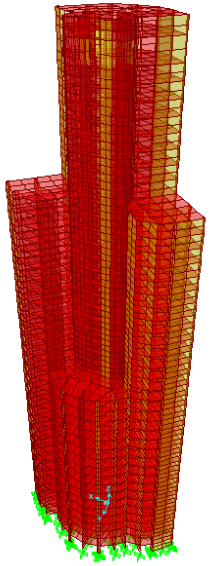
$$Kx = F$$

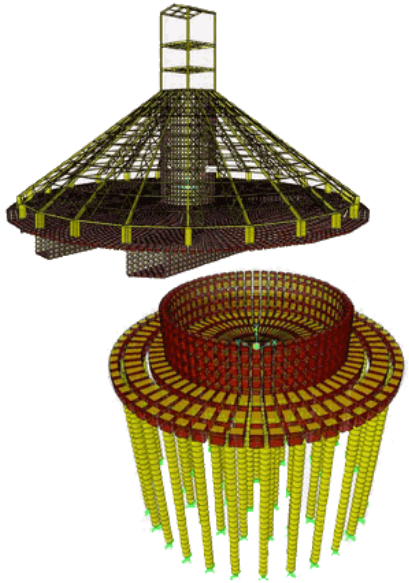
  - $K \in \mathbb{R}^{n \times n}$: "stiffness matrix"
  - $F \in \mathbb{R}^{n}$: "load vector"
  - $x \in \mathbb{R}^{n}$: "displacement vector"

Example: Structural Analysis

- It is common engineering practice to use Hooke's law to simulate complex structures, which leads to large linear systems







(from SAP2000, structural analysis software)

Example: Economics

- Leontief awarded Nobel Prize in Economics in 1973 for developing a linear input/output model for production/consumption of goods
- Consider an economy in which n goods are produced and consumed
  - $A \in \mathbb{R}^{n \times n}$: a_{ij} represents the amount of good j required to produce a unit of good i
  - $x \in \mathbb{R}^n$: x_i is number of units of good i produced
  - $d \in \mathbb{R}^n$: d_i is consumer demand for good i
- In general $a_{ii} = 0$, and A may be sparse

Example: Economics

- The total amount of x_i produced is given by the sum of consumer demand d_i and the amount of x_i required to produce each x_j

$$x_i = \underbrace{a_{i1}x_1 + a_{i2}x_2 + \cdots + a_{in}x_n}_{\text{production of other goods}} + d_i$$

- Hence $x = Ax + d$ or,

$$(\mathrm{I} - A)x = d$$

- Solve for x to determine the required amount of production of each good
- If we consider many goods (e.g. an entire economy), then we get a large linear system
- Can be used to predict the effect of disruptions in the supply chain

Summary

- Matrix computations are very common
- Numerical Linear Algebra provides us with a toolbox
  for performing these computations in an efficient and stable manner
- In most cases, we can use these tools as a black box,
  but it's important to understand what they do
  - pick the right algorithm for a given situation
    (e.g. exploit structure of a problem: symmetry, sparsity, etc)
  - understand how and when the algorithm fail

Preliminaries

Preliminaries

- In this section we will focus on linear systems

$$Ax = b$$

  with matrix $A \in \mathbb{R}^{n \times n}$, unknown vector $x \in \mathbb{R}^n$
  and the right-hand side vector $b \in \mathbb{R}^n$
- Recall that it is often helpful to think of matrix multiplication
  as a linear combination of the columns of A, where x_j are the coefficients
- That is, we have

$$Ax = \sum_{j=1}^{n} x_j a_{(:,j)}$$

  where $a_{(:,j)} \in \mathbb{R}^n$ is the j-th column of A and x_j are scalars

Preliminaries

- This can be displayed schematically as

$$Ax = \left[\begin{array}{c|c|c|c} & & & \\ a_{(:,1)} & a_{(:,2)} & \cdots & a_{(:,n)} \\ & & & \end{array}\right] \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} =$$

$$= \quad x_1 \begin{bmatrix} \\ a_{(:,1)} \\ \\ \end{bmatrix} + \cdots + \quad x_n \begin{bmatrix} \\ a_{(:,n)} \\ \\ \end{bmatrix}$$

Preliminaries

- We therefore interpret $Ax = b$ as:
  "x is the vector of coordinates of b in the basis of columns of A"
- Often this is a more helpful point of view than conventional interpretation of "dot-product of matrix row with vector"
- Now we see that $Ax = b$ has a solution if

$$b \in \text{span}\{a_{(:,1)}, a_{(:,2)}, \cdots, a_{(:,n)}\}$$

  (this holds even for a non-square A)
- Denote

$$\text{image}(A) = \text{span}\{a_{(:,1)}, a_{(:,2)}, \cdots, a_{(:,n)}\}$$

Preliminaries

Existence and Uniqueness

- If $b \in \text{image}(A)$, then solution $x \in \mathbb{R}^n$ exists
  - if solution x exists and the columns $\{a_{(:,1)}, a_{(:,2)}, \cdots, a_{(:,n)}\}$ are linearly independent, then x is unique
    (if x and y are both solutions, then $A(x-y) = 0$, therefore $x = y$)
  - if x is a solution and $z \neq 0$ is such that $Az = 0$,
    then also $A(x + \gamma z) = b$ for any $\gamma \in \mathbb{R}$,
    so there are infinitely many solutions
- If $b \notin \text{image}(A)$ then $Ax = b$ has no solution

Preliminaries

- The inverse map $A^{-1} : \mathbb{R}^n \to \mathbb{R}^n$ is well-defined if and only if $Ax = b$ has unique solution for any $b \in \mathbb{R}^n$
- The inverse matrix $A^{-1} \in \mathbb{R}^{n \times n}$ such that $AA^{-1} = A^{-1}A = I$ exists if any of the following equivalent conditions are satisfied
  - $\det(A) \neq 0$
  - $\operatorname{rank}(A) = n$
  - $Az \neq 0$ for any $z \neq 0$ (null space of A is $\{0\}$)
- A is non-singular if A^{-1} exists, and then $x = A^{-1}b \in \mathbb{R}^n$
- A is singular if A^{-1} does not exist

Norms

- A norm $\|\cdot\| : V \to \mathbb{R}$ is a function on a vector space V that satisfies
  - positive definiteness, $\|x\| \geq 0$ and $\|x\| = 0 \implies x = 0$
  - absolute homogeneity, $\|\gamma x\| = |\gamma|\|x\|$, for $\gamma \in \mathbb{R}$
  - triangle inequality, $\|x + y\| \leq \|x\| + \|y\|$

Norms

- The triangle inequality implies another helpful inequality: the "reverse triangle inequality"

$$\big|\|x\| - \|y\|\big| \leq \|x - y\|$$

- Proof:

$$\|x\| = \|(x-y)+y\| \leq \|x-y\| + \|y\| \implies \|x\| - \|y\| \leq \|x-y\|$$
$$\|y\| = \|(y-x)+x\| \leq \|y-x\| + \|x\| \implies \|y\| - \|x\| \leq \|x-y\|$$

- Therefore $\big|\|x\| - \|y\|\big| \leq \|x - y\|$

Vector Norms

- Let's now introduce some common norms on $\mathbb{R}^n$
- Most common norm is the Euclidean norm (or 2-norm):

$$\|x\|_2 = \sqrt{\sum_{j=1}^n x_j^2}$$

- 2-norm is special case of the p-norm for any $p \geq 1$:

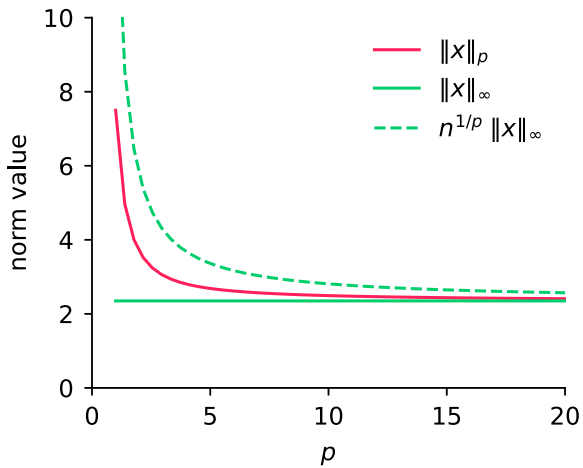
$$\|x\|_p = \left(\sum_{j=1}^n |x_j|^p\right)^{1/p}$$

- Condition $p \geq 1$ is required for the triangle inequality
- Norm $\|x\|_p$ approaches $\|x\|_\infty$ as $p \to \infty$

$$\|x\|_\infty = \max_{1 \leq i \leq n} |x_i|$$

Example: Limit of p-norm

- See [examples/unit2/norm_inf.py]
- For vector $x = (1.2, 0.5, -0.1, 2.3, -1.05, -2.35)^T \in \mathbb{R}^6$
- $\|x\|_\infty = 2.35$ (component of x with the largest magnitude)
- Norm $\|x\|_p$ approaches $\|x\|_\infty$ as $p \to \infty$
- Bounds: $\|x\|_\infty \leq \|x\|_p \leq n^{1/p}\|x\|_\infty$



Vector Norms

- We generally use whichever norm is most convenient/appropriate for a given problem, e.g. 2-norm for least-squares analysis
- Different norms give different (but related) measures of size
- An important fact is:

  All norms on a finite dimensional space (such as $\mathbb{R}^n$) are equivalent

Vector Norms

- That is, let $\|\cdot\|_a$ and $\|\cdot\|_b$ be two norms on a finite dimensional space V, then $\exists\ c_1, c_2 > 0$ such that for any $x \in V$

$$c_1\|x\|_a \leq \|x\|_b \leq c_2\|x\|_a$$

- Also, from above we have $\frac{1}{c_2}\|x\|_b \leq \|x\|_a \leq \frac{1}{c_1}\|x\|_b$
- Hence if we can derive an inequality in one norm on V, it applies (after appropriate scaling) in any other norm as well

Vector Norms

- Norm $\|x\|_2$ bounds norm $\|x\|_1$

$$\|x\|_2 \leq \|x\|_1 \leq \sqrt{n}\|x\|_2$$

- Proof of $\|x\|_2 \leq \|x\|_1$

$$\|x\|_1^2 = \left(\sum_{i=1}^n |x_i|\right)^2 = \left(\sum_{i=1}^n |x_i|\right)\left(\sum_{j=1}^n |x_j|\right) =$$
$$= \sum_{i=1}^n \sum_{j=1}^n |x_i|\,|x_j| \geq \sum_{i=1}^n |x_i|\,|x_i| = \sum_{i=1}^n |x_i|^2 = \|x\|_2^2$$

- Proof of $\|x\|_1 \leq \sqrt{n}\|x\|_2$. The Cauchy-Schwarz inequality

$$\sum_{i=1}^n a_i b_i \leq \left(\sum_{i=1}^n a_i^2\right)^{1/2}\left(\sum_{i=1}^n b_i^2\right)^{1/2}$$

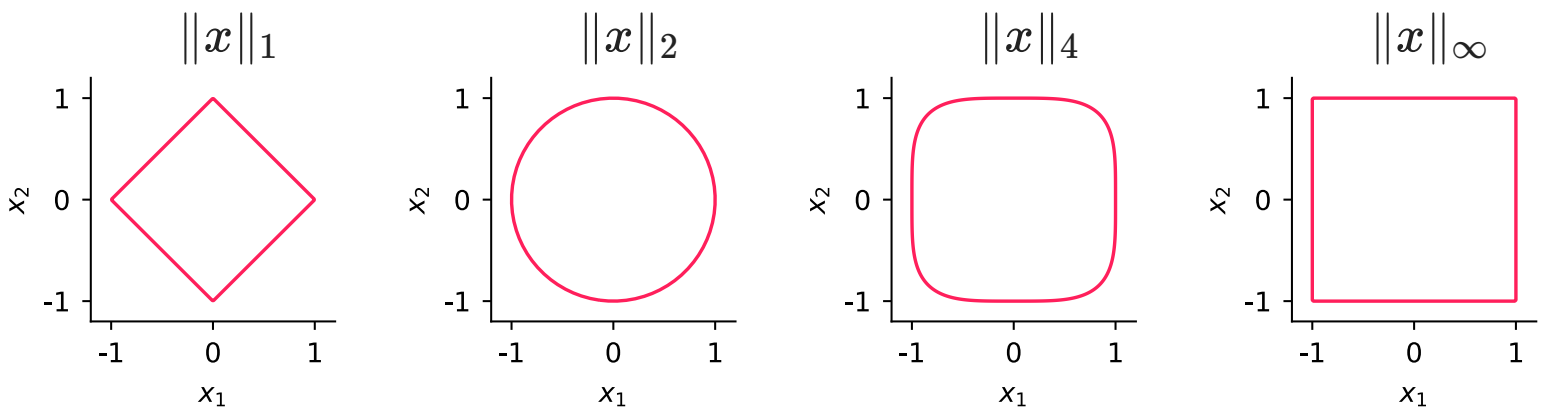
with $a_i = 1$ and $b_i = |x_i|$ gives

$$\|x\|_1 = \sum_{i=1}^n 1\,|x_i| \leq \left(\sum_{i=1}^n 1^2\right)^{1/2}\left(\sum_{i=1} |x_i|^2\right)^{1/2} = \sqrt{n}\,\|x\|_2$$

Vector Norms

- Each norm produces a different unit circle

$$\{x \in \mathbb{R}^2 : \|x\|_p = 1\}$$



- Norm $\|x\|_p$ approaches $\|x\|_\infty$ as $p \to \infty$
- Commonly used norms are $\|x\|_1$, $\|x\|_2$, and $\|x\|_\infty$

Matrix Norms

- There are many ways to define norms on matrices
- For example, the Frobenius norm is defined as

$$\|A\|_F = \left(\sum_{i=1}^{n} \sum_{j=1}^{n} |a_{ij}|^2 \right)^{1/2}$$

- If we think of A as a vector in $\mathbb{R}^{n^2}$,
  then Frobenius is equivalent to the vector 2-norm of A

Matrix Norms

- Matrix norms induced by vector norms are most useful
- Here, matrix p-norm is induced by vector p-norm

$$\|A\|_p = \max_{x \neq 0} \frac{\|Ax\|_p}{\|x\|_p} = \max_{\|x\|_p = 1} \|Ax\|_p$$

- This definition implies the useful property

$$\|Ax\|_p \leq \|A\|_p \|x\|_p$$

since

$$\|Ax\|_p = \frac{\|Ax\|_p}{\|x\|_p} \|x\|_p \leq \left(\max_{v \neq 0} \frac{\|Av\|_p}{\|v\|_p} \right) \|x\|_p = \|A\|_p \|x\|_p$$

Matrix Norms

- The 1-norm and ∞-norm can be calculated straightforwardly:

$$\|A\|_1 = \max_{1 \le j \le n} \|a_{(:,j)}\|_1 \qquad \text{(max column sum)}$$

$$\|A\|_\infty = \max_{1 \le i \le n} \|a_{(i,:)}\|_1 \qquad \text{(max row sum)}$$

- Later we will see how to compute the 2-norm of a matrix

Example: Matrix Norm Using Monte Carlo

- How to compute the matrix norm induced by a "black box" vector norm?
- One approach is the Monte-Carlo method
  that solves problems using repeated random sampling
- Recall the definition of a matrix norm induced by vector norm

$$\|A\| = \max_{x \neq 0} \frac{\|Ax\|}{\|x\|}$$

- See [examples/unit2/norm_monte_carlo.py]
- Warning: Common norms can be computed with more efficient methods!

Condition Number

- Recall from Unit 0 that the condition number of $A \in \mathbb{R}^{n \times n}$ is defined as

$$\kappa(A) = \|A\| \, \|A^{-1}\|$$

- The value of $\kappa(A)$ depends on which norm we use
- `numpy.linalg.cond` computes the condition number for various norms
- If A is a singular square matrix, then by convention $\kappa(A) = \infty$

Residual

- Recall that the residual $r(x) = b - Ax$
  was crucial in least-squares problems
- It is also crucial in assessing the accuracy
  of a proposed solution $(\hat{x})$ to a linear system $Ax = b$
- Key point: The residual $r(\hat{x})$ is straightforward to compute,
  while the error $\Delta x = x - \hat{x}$ is not (without knowing the exact solution)

Residual

- We have that $\|\Delta x\| = \|x - \hat{x}\| = 0$ if and only if $\|r(\hat{x})\| = 0$
- However, small residual doesn't necessarily imply small $\|\Delta x\|$
- Observe that

$$\|\Delta x\| = \|x - \hat{x}\| = \|A^{-1}(b - A\hat{x})\| = \|A^{-1}r(\hat{x})\| \leq \|A^{-1}\|\|r(\hat{x})\|$$

Hence

$$\frac{\|\Delta x\|}{\|\hat{x}\|} \leq \frac{\|A^{-1}\|\|r(\hat{x})\|}{\|\hat{x}\|} = \frac{\|A\|\|A^{-1}\|\|r(\hat{x})\|}{\|A\|\|\hat{x}\|} = \kappa(A)\frac{\|r(\hat{x})\|}{\|A\|\|\hat{x}\|} \qquad (*)$$

Residual

- Define the relative residual as

$$\frac{\|r(\hat{x})\|}{\|A\|\|\hat{x}\|}$$

- Then our inequality $(*)$ states that
  "relative error is bounded by condition number times the relative residual"
- This is just like our condition number relationship from Unit 0:

$$\kappa(A) \geq \frac{\|\Delta x\|/\|x\|}{\|\Delta b\|/\|b\|}, \qquad \text{i.e.} \qquad \frac{\|\Delta x\|}{\|x\|} \leq \kappa(A)\frac{\|\Delta b\|}{\|b\|} \qquad (**)$$

- The reason $(*)$ and $(**)$ are related is that
  the residual measures the input pertubation (Δb) in $Ax = b$
- To see this, let's consider $Ax = b$ to be a map $b \in \mathbb{R}^n \to x \in \mathbb{R}^n$

Residual

- Then we can consider $\hat{x}$ to be the exact solution for some perturbed input $\hat{b} = b + \Delta b$

$$A\hat{x} = \hat{b}$$

- The residual associated with $\hat{x}$ is

$$r(\hat{x}) = b - A\hat{x} = b - \hat{b} = -\Delta b$$

  i.e. $\|r(\hat{x})\| = \|\Delta b\|$
- In general, a (backward) stable algorithm gives us the exact solution to a slightly perturbed problem, i.e. a small residual
- This is a reasonable expectation for a stable algorithm: rounding error doesn't accumulate, so effective input perturbation is small

Example: Residual vs. Error

- From Heath's book (Example 2.8)
- Consider a 2×2 example to clearly demonstrate the difference between residual and error

$$Ax = \begin{bmatrix} 0.913 & 0.659 \\ 0.457 & 0.330 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0.254 \\ 0.127 \end{bmatrix} = b$$

- The exact solution is given by $x = [1, -1]^T$
- Suppose we compute two different approximate solutions

$$\hat{x}^{(1)} = \begin{bmatrix} -0.0827 \\ 0.5 \end{bmatrix}, \qquad \hat{x}^{(2)} = \begin{bmatrix} 0.999 \\ -1.001 \end{bmatrix}$$

Example: Residual vs. Error

- Then,

$$\|r(\hat{x}^{(1)})\|_1 = 2.1 \times 10^{-4}, \qquad \|r(\hat{x}^{(2)})\|_1 = 2.4 \times 10^{-2}$$

- but

$$\|x - \hat{x}^{(1)}\|_1 = 2.58, \qquad \|x - \hat{x}^{(2)}\|_1 = 0.002$$

- In this case, $\hat{x}^{(2)}$ is better solution, but has larger residual!
- This is possible here because $\kappa(A) = 1.25 \times 10^4$ is quite large (relative error $\leq 1.25 \times 10^4 \times$ relative residual)

Solving $Ax = b$

Solving $Ax = b$

- Familiar idea for solving $Ax = b$ is to use Gaussian elimination to transform $Ax = b$ to a triangular system
- What is a triangular system?
  - upper triangular $U \in \mathbb{R}^{n \times n}$

$$u_{ij} = 0 \text{ for } i > j$$

$$U = \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{bmatrix}$$

  - lower triangular $L \in \mathbb{R}^{n \times n}$

$$\ell_{ij} = 0 \text{ for } i < j$$

$$L = \begin{bmatrix} l_{11} & 0 & 0 \\ l_{21} & l_{22} & 0 \\ l_{31} & l_{32} & l_{33} \end{bmatrix}$$

- Question: Why triangular?
- Answer: Because triangular systems are easy to solve!

Solving $Ax = b$

- For an upper-triangular system $Ux = b$, we can use backward substitution

$$
\begin{aligned}
x_n &= b_n / u_{nn} \\
x_{n-1} &= (b_{n-1} - u_{n-1,n} x_n) / u_{n-1,n-1} \\
&\cdots \\
x_j &= \left(b_j - \textstyle\sum_{k=j+1}^{n} u_{jk} x_k \right) / u_{jj}
\end{aligned}
$$

Solving $Ax = b$

- For a lower triangular system $Lx = b$, we can use forward substitution

$$
\begin{aligned}
x_1 &= b_1/\ell_{11} \\
x_2 &= (b_2 - \ell_{21}x_1)/\ell_{22} \\
&\dots \\
x_j &= \left(b_j - \textstyle\sum_{k=1}^{j-1} \ell_{jk}x_k\right)/\ell_{jj}
\end{aligned}
$$

Asymptotic Notation

- To simplify the cost estimation for an algorithm, we analyze its asymptotic behavior as the size of the problem increases ($n \to \infty$)
- Notation $f(n) \sim g(n)$ refers to asymptotic equivalence

$$\lim_{n \to \infty} \frac{f(n)}{g(n)} = 1$$

- Notation $f(n) = \mathcal{O}(g(n))$ refers to an asymptotic upper bound

$$|f(n)| \leq M|g(n)|$$

  for all $n \geq N$, where $M > 0$ and $N > 0$
- If $f(n) \sim g(n)$, then $f(n) = \mathcal{O}(g(n))$. The opposite is not true!
- We prefer "$\sim$" since it indicates the scaling factor of the leading term
- For example, if $f(n) = n^2/4 + n$, then $f(n) = \mathcal{O}(n^2)$, whereas $f(n) \sim n^2/4$

Solving $Ax = b$

- Backward (and forward) substitution
  can be implemented with a double nested loop
- It requires just one pass through the matrix!
- The computational work is dominated by evaluating the sum

$$\sum_{k=1}^{j-1} \ell_{jk} x_k \quad j = 1, \ldots, n$$

  which takes $j - 1$ additions and multiplications for each j
- So the total number of floating point operations is asymptotically

$$2 \sum_{j=1}^{n} j = \frac{2n(n+1)}{2} \sim n^2$$

Solving $Ax = b$

- How can we transform $Ax = b$ to a triangular system?
- Observation: If we multiply $Ax = b$ by a nonsingular matrix M, then the new system $MAx = Mb$ has the same solution
- We can devise a sequence of matrices

$$M_1, M_2, \ldots, M_{n-1}$$

  such that $M = M_{n-1} \ldots M_1$ and $U = MA$ is upper triangular
- Gaussian elimination provides such a sequence and gives the transformed system $Ux = Mb$

LU Factorization

- We will show shortly that if $MA = U$,
  then $L = M^{-1}$ is lower triangular
- Therefore, we obtain that the matrix factorizes into

$$A = M^{-1}U = LU$$

  a product of lower (L) and upper (U) triangular matrices
- This is the LU factorization of A

LU Factorization

- LU factorization is a common way of solving linear systems!
- Once a factorization $A = LU$ is known, the system

$$LUx = b$$

  is solved in two steps
  - lower triangular: $Ly = b$
  - upper triangular: $Ux = y$

LU Factorization

- Next question: How can we find $M_1, M_2, \cdots, M_{n-1}$?
- We need to be able to annihilate selected entries of A below the diagonal in order to obtain an upper-triangular matrix
- To do this, we use elementary elimination matrices
- Let L_j denote j-th elimination matrix
- From now on, we denote them L_j rather than M_j since elimination matrices are lower triangular

LU Factorization

- Here we describe how to proceed from step $j-1$ to step j
- Let $X = L_{j-1}L_{j-2}\cdots L_1 A$ denote the matrix at the start of step j, and $x_{(:,k)} \in \mathbb{R}^n$ denote column k of X

$$X = \begin{bmatrix} x_{11} & \cdots & x_{1,j-1} & x_{1j} & x_{1,j+1} & \cdots & x_{1n} \\ \vdots & \ddots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & \cdots & x_{j-1,j-1} & x_{j-1,j} & x_{j-1,j+1} & \cdots & x_{j-1,n} \\ 0 & \cdots & 0 & x_{jj} & x_{j,j+1} & \cdots & x_{jn} \\ 0 & \cdots & 0 & x_{j+1,j} & x_{j+1,j+1} & \cdots & x_{j+1,n} \\ \vdots & \ddots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & \cdots & 0 & x_{nj} & x_{n,j+1} & \cdots & x_{nn} \end{bmatrix}$$

LU Factorization

- We are looking for a matrix L_j such that multiplication $L_j X$
  - eliminates elements below the diagonal in $x_{(:,j)}$
  - does not modify columns $x_{(:,k)}$ for $k = 1, \ldots, j-1$
- Let's define L_j such that

$$L_j x_{(:,j)} = \begin{bmatrix} 1 & \cdots & 0 & 0 & \cdots & 0 \\ \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ 0 & \cdots & 1 & 0 & \cdots & 0 \\ 0 & \cdots & -x_{j+1,j}/x_{jj} & 1 & \cdots & 0 \\ \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ 0 & \cdots & -x_{nj}/x_{jj} & 0 & \cdots & 1 \end{bmatrix} \begin{bmatrix} x_{1j} \\ \vdots \\ x_{jj} \\ x_{j+1,j} \\ \vdots \\ x_{nj} \end{bmatrix} = \begin{bmatrix} x_{1j} \\ \vdots \\ x_{jj} \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

LU Factorization

- For brevity, we denote $\ell_{ij} = x_{ij}/x_{jj}$ and define

$$L_j = \begin{bmatrix} 1 & \cdots & 0 & 0 & \cdots & 0 \\ \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ 0 & \cdots & 1 & 0 & \cdots & 0 \\ 0 & \cdots & -\ell_{j+1,j} & 1 & \cdots & 0 \\ \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ 0 & \cdots & -\ell_{nj} & 0 & \cdots & 1 \end{bmatrix}$$

LU Factorization

- Using elementary elimination matrices,
  we can reduce A to an upper triangular form, one column at a time
- Schematically, for a 4×4 matrix, we have

$$\begin{bmatrix} * & * & * & * \\ * & * & * & * \\ * & * & * & * \\ * & * & * & * \end{bmatrix} \xrightarrow{L_1} \begin{bmatrix} * & * & * & * \\ 0 & * & * & * \\ 0 & * & * & * \\ 0 & * & * & * \end{bmatrix} \xrightarrow{L_2} \begin{bmatrix} * & * & * & * \\ 0 & * & * & * \\ 0 & 0 & * & * \\ 0 & 0 & * & * \end{bmatrix}$$

$$A \qquad\qquad L_1 A \qquad\qquad L_2 L_1 A$$

- Key point: L_j does not modify columns $1, \ldots, j-1$ of $L_{j-1} L_{j-2} \cdots L_1 A$

LU Factorization

- After $n-1$ steps, we obtain an upper triangular matrix

$$U = L_{n-1} \cdots L_2 L_1 A = \begin{bmatrix} * & * & * & * \\ 0 & * & * & * \\ 0 & 0 & * & * \\ 0 & 0 & 0 & * \end{bmatrix}$$

LU Factorization

- We have $L_{n-1} \cdots L_2 L_1 A = U$
- To form a factorization $A = LU$,
  we need $L = (L_{n-1} \cdots L_2 L_1)^{-1} = L_1^{-1} L_2^{-1} \cdots L_{n-1}^{-1}$
- First observation:
  L_j^{-1} is obtained by negating the subdiagonal elements of L_j

$$L_j = \begin{bmatrix} 1 & \cdots & 0 & 0 & \cdots & 0 \\ \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ 0 & \cdots & 1 & 0 & \cdots & 0 \\ 0 & \cdots & -\ell_{j+1,j} & 1 & \cdots & 0 \\ \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ 0 & \cdots & -\ell_{nj} & 0 & \cdots & 1 \end{bmatrix} \qquad L_j^{-1} = \begin{bmatrix} 1 & \cdots & 0 & 0 & \cdots & 0 \\ \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ 0 & \cdots & 1 & 0 & \cdots & 0 \\ 0 & \cdots & \ell_{j+1,j} & 1 & \cdots & 0 \\ \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ 0 & \cdots & \ell_{nj} & 0 & \cdots & 1 \end{bmatrix}$$

LU Factorization

- $L_j L_j^{-1} = I$ can be verified directly by multiplication
- Intuitive explanation
  - multiplication $L_j v$ subtracts a scaled component v_j
  - so the inverse should add it back (v_j itself is unchanged)

$$L_j \begin{bmatrix} v_1 \\ \vdots \\ v_j \\ v_{j+1} \\ \vdots \\ v_n \end{bmatrix} = \begin{bmatrix} v_1 \\ \vdots \\ v_j \\ v_{j+1} - \ell_{j+1,j} v_j \\ \vdots \\ v_n - \ell_{nj} v_j \end{bmatrix}$$

$$L_j^{-1} \begin{bmatrix} v_1 \\ \vdots \\ v_j \\ v_{j+1} \\ \vdots \\ v_n \end{bmatrix} = \begin{bmatrix} v_1 \\ \vdots \\ v_j \\ v_{j+1} + \ell_{j+1,j} v_j \\ \vdots \\ v_n + \ell_{nj} v_j \end{bmatrix}$$

LU Factorization

- Second observation: consider $L_{j-1}^{-1} L_j^{-1}$

$$\underbrace{\begin{bmatrix} 1 & \cdots & 0 & 0 & 0 & \cdots & 0 \\ \vdots & \ddots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & \cdots & 1 & 0 & 0 & \cdots & 0 \\ 0 & \cdots & \ell_{j,j-1} & 1 & 0 & \cdots & 0 \\ 0 & \cdots & \ell_{j+1,j-1} & 0 & 1 & \cdots & 0 \\ \vdots & \ddots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & \cdots & \ell_{n,j-1} & 0 & 0 & \cdots & 1 \end{bmatrix}}_{L_{j-1}^{-1}} \underbrace{\begin{bmatrix} 1 & \cdots & 0 & 0 & 0 & \cdots & 0 \\ \vdots & \ddots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & \cdots & 1 & 0 & 0 & \cdots & 0 \\ 0 & \cdots & 0 & 1 & 0 & \cdots & 0 \\ 0 & \cdots & 0 & \ell_{j+1,j} & 1 & \cdots & 0 \\ \vdots & \ddots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & \cdots & 0 & \ell_{nj} & 0 & \cdots & 1 \end{bmatrix}}_{L_j^{-1}} = \begin{bmatrix} 1 & \cdots & 0 & 0 & 0 & \cdots & 0 \\ \vdots & \ddots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & \cdots & 1 & 0 & 0 & \cdots & 0 \\ 0 & \cdots & \ell_{j,j-1} & 1 & 0 & \cdots & 0 \\ 0 & \cdots & \ell_{j+1,j-1} & \ell_{j+1,j} & 1 & \cdots & 0 \\ \vdots & \ddots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & \cdots & \ell_{n,j-1} & \ell_{nj} & 0 & \cdots & 1 \end{bmatrix}$$

LU Factorization

- Therefore, by generalizing to all $n-1$ matrices

$$L = L_1^{-1} L_2^{-1} \cdots L_{n-1}^{-1} = \begin{bmatrix} 1 & & & & \\ \ell_{21} & 1 & & & \\ \ell_{31} & \ell_{32} & 1 & & \\ \vdots & \vdots & \ddots & \ddots & \\ \ell_{n1} & \ell_{n2} & \cdots & \ell_{n,n-1} & 1 \end{bmatrix}$$

- So we simply collect the subdiagonal terms from all steps of factorization

LU Factorization

- Therefore, basic LU factorization algorithm is

```
1: U = A, L = I
2: for j = 1 : n − 1 do
3:     for i = j + 1 : n do
4:         ℓ_ij = u_ij / u_jj
5:         for k = j : n do
6:             u_ik = u_ik − ℓ_ij u_jk
7:         end for
8:     end for
9: end for
```

- Note that the entries of U are updated each iteration so at the start of step j, $U = L_{j-1} L_{j-2} \cdots L_1 A$
- Here line 4 comes straight from the definition $\ell_{ij} = \frac{u_{ij}}{u_{jj}}$

LU Factorization

- Line 6 accounts for the effect of L_j on columns $k = j, \ldots, n$ of U
- For $k = j : n$ we have

$$L_j u_{(:,k)} = \begin{bmatrix} 1 & \cdots & 0 & 0 & \cdots & 0 \\ \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ 0 & \cdots & 1 & 0 & \cdots & 0 \\ 0 & \cdots & -\ell_{j+1,j} & 1 & \cdots & 0 \\ \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ 0 & \cdots & -\ell_{nj} & 0 & \cdots & 1 \end{bmatrix} \begin{bmatrix} u_{1k} \\ \vdots \\ u_{jk} \\ u_{j+1,k} \\ \vdots \\ u_{nk} \end{bmatrix} = \begin{bmatrix} u_{1k} \\ \vdots \\ u_{jk} \\ u_{j+1,k} - \ell_{j+1,j} u_{jk} \\ \vdots \\ u_{nk} - \ell_{nj} u_{jk} \end{bmatrix}$$

- The right hand side is the updated k-th column of U, which is computed in line 6

LU Factorization

- LU factorization involves a triple nested loop, hence $\mathcal{O}(n^3)$ operations
- Careful operation counting shows LU factorization requires
  - $\sim \frac{1}{3}n^3$ additions
  - $\sim \frac{1}{3}n^3$ multiplications
- Therefore $\sim \frac{2}{3}n^3$ operations in total

Solving Linear Systems Using LU

- To solve $Ax = b$, we perform the following three steps:
  - Step 1: Factorize A into $A = LU$: $\sim \frac{2}{3}n^3$
  - Step 2: Solve $Ly = b$ by forward substitution: $\sim n^2$
  - Step 3: Solve $Ux = y$ by backward substitution: $\sim n^2$
- The total work, dominated by Step 1, is $\sim \frac{2}{3}n^3$

Solving Linear Systems Using LU

- An alternative approach would be to first compute A^{-1} and evaluate $x = A^{-1}b$, but this is a bad idea!
- Question: How would we compute A^{-1}?

Solving Linear Systems Using LU

- Answer: Let $a^{\text{inv}}_{(:,k)}$ denote the k-th column of A^{-1}, then $a^{\text{inv}}_{(:,k)}$ must satisfy

$$Aa^{\text{inv}}_{(:,k)} = e_k$$

  where e_k is the k-th basis vector
- Therefore, inverting matrix A reduces to solving $Ax = b$ for n various b
- We first factorize $A = LU$, then forward/backward substitute for

$$LUa^{\text{inv}}_{(:,k)} = e_k, \quad k = 1, \ldots, n$$

Solving Linear Systems Using LU

- Solving linear systems using A^{-1} is inefficient!
  - one pair of substitutions requires $\sim 2n^2$ operations
  - n pairs of substitutions require $\sim 2n^3$ operations
  - evaluating $A^{-1}b$ takes $\sim 2n^2$ operations
    (as many as one pair of substitutions)
- A rule of thumb in Numerical Linear Algebra:
  It is rarely a good idea to compute A^{-1} explicitly

Solving Linear Systems Using LU

- Another case where LU factorization is very helpful is if we want to solve $Ax = b_i$ for several different right-hand sides b_i, $i = 1, \ldots, k$
- We incur the $\sim \frac{2}{3}n^3$ cost only once, and then each subsequent pair of forward/backward substitutions costs only $\sim 2n^2$
- Makes a huge difference if n is large!

Stability of Gaussian Elimination

- There is a problem with the LU algorithm presented above
- Consider the matrix

$$A = \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix}$$

- A is nonsingular, well-conditioned ($\kappa(A) \approx 2.62$)
  but LU factorization fails at first step (division by zero)

Stability of Gaussian Elimination

- LU factorization doesn't fail for

$$A = \begin{bmatrix} 10^{-20} & 1 \\ 1 & 1 \end{bmatrix}$$

but we get

$$L = \begin{bmatrix} 1 & 0 \\ 10^{20} & 1 \end{bmatrix}, \qquad U = \begin{bmatrix} 10^{-20} & 1 \\ 0 & 1 - 10^{20} \end{bmatrix}$$

Stability of Gaussian Elimination

- Let's suppose that $-10^{20} \in \mathbb{F}$ (a floating point number) and that $\text{round}(1 - 10^{20}) = -10^{20}$
- Then in finite precision arithmetic we get

$$\widetilde{L} = \begin{bmatrix} 1 & 0 \\ 10^{20} & 1 \end{bmatrix}, \qquad \widetilde{U} = \begin{bmatrix} 10^{-20} & 1 \\ 0 & -10^{20} \end{bmatrix}$$

Stability of Gaussian Elimination

- Hence due to rounding error we obtain

$$\widetilde{L}\widetilde{U} = \begin{bmatrix} 10^{-20} & 1 \\ 1 & 0 \end{bmatrix}$$

  which is not close to

$$A = \begin{bmatrix} 10^{-20} & 1 \\ 1 & 1 \end{bmatrix}$$

- Then, for example, let $b = [3, 3]^T$
  - using $\widetilde{L}\widetilde{U}$, we get $\tilde{x} = [3, 3]^T$
  - true answer is $x = [0, 3]^T$
- The relative error is large
  even though the problem is well-conditioned

Stability of Gaussian Elimination

- In this example, standard Gaussian elimination yields a large residual
- Or equivalently, it yields the exact solution to a problem corresponding to a large input perturbation: $\Delta b = [0, 3]^T$
- So the algorithm is unstable!
- In this case the cause of the large error in x is numerical instability, not ill-conditioning
- To stabilize Gaussian elimination, we need to permute rows, i.e. perform pivoting

Pivoting

- Recall the Gaussian elimination process

$$\begin{bmatrix} * & * & * & * \\ & x_{jj} & * & * \\ & * & * & * \\ & * & * & * \end{bmatrix} \longrightarrow \begin{bmatrix} * & * & * & * \\ & x_{jj} & * & * \\ & 0 & * & * \\ & 0 & * & * \end{bmatrix}$$

- But we could just as easily do

$$\begin{bmatrix} * & * & * & * \\ & * & * & * \\ & x_{ij} & * & * \\ & * & * & * \end{bmatrix} \longrightarrow \begin{bmatrix} * & * & * & * \\ & 0 & * & * \\ & x_{ij} & * & * \\ & 0 & * & * \end{bmatrix}$$

Partial Pivoting

- The entry x_{ij} is called the pivot, and flexibility in choosing the pivot is essential otherwise we can't deal with:

$$A = \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix}$$

- Choosing the pivot as the largest element in column j improves numerical stability. This is called partial pivoting
- Full pivoting additionally permutes the columns and looks for the largest over $\mathcal{O}(n^2)$ elements, which is costly and only marginally beneficial for stability
- This ensures that each ℓ_{ij} entry — which acts as a multiplier in the LU factorization process — satisfies $|\ell_{ij}| \leq 1$

Partial Pivoting

- To maintain the triangular LU structure,
  we permute rows by premultiplying by permutation matrices

$$\underset{\text{pivot selection}}{\begin{bmatrix} * & * & * & * \\ & * & * & * \\ & * & * & * \\ & x_{ij} & * & * \end{bmatrix}} \xrightarrow{P_1} \underset{\text{row swap}}{\begin{bmatrix} * & * & * & * \\ & x_{ij} & * & * \\ & * & * & * \\ & * & * & * \end{bmatrix}} \xrightarrow{L_1} \begin{bmatrix} * & * & * & * \\ & x_{ij} & * & * \\ & 0 & * & * \\ & 0 & * & * \end{bmatrix}$$

- In this case

$$P_1 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}$$

and each P_j is obtained by swapping two rows of I

Partial Pivoting

- Therefore, with partial pivoting we obtain

$$L_{n-1}P_{n-1}\cdots L_2P_2L_1P_1A = U$$

- It can be shown (we omit the details here, see Trefethen & Bau) that this can be rewritten as

$$PA = LU$$

  where $P = P_{n-1}\cdots P_2P_1$. Note that L is not the same as without pivoting
- Theorem: Gaussian elimination with partial pivoting produces nonsingular factors L and U if and only if A is nonsingular

Partial Pivoting

- Pseudocode for LU factorization with partial pivoting (new code is highlighted):

```
U = A, L = I, P = I
for j = 1 : n − 1 do
    Select i(≥ j) that maximizes |u_ij|
    Swap rows of U: u_(j,j:n) ↔ u_(i,j:n)
    Swap rows of L: ℓ_(j,1:j−1) ↔ ℓ_(i,1:j−1)
    Swap rows of P: p_(j,:) ↔ p_(i,:)
    for i = j + 1 : n do
        ℓ_ij = u_ij / u_jj
        for k = j : n do
            u_ik = u_ik − ℓ_ij u_jk
        end for
    end for
end for
```

- Again this requires $\sim \frac{2}{3}n^3$ floating point operations

Partial Pivoting: Solve $Ax = b$

- To solve $Ax = b$ using the factorization $PA = LU$
  - Multiply through by P to obtain $PAx = LUx = Pb$
  - Solve $Ly = Pb$ using forward substitution
  - Then solve $Ux = y$ using back substitution

Partial Pivoting in Python

- Python's `scipy.linalg.lu` function can do LU factorization with pivoting

```
>>> import numpy as np
>>> import scipy.linalg
>>> A=np.random.rand(4, 4)
>>> (P,L,U) = scipy.linalg.lu(A)
>>> A
array([[0.48657354, 0.72177328, 0.89725033, 0.10555858],
       [0.19356039, 0.21192135, 0.001038  , 0.20308355],
       [0.04709362, 0.82519218, 0.29700521, 0.85089909],
       [0.35533098, 0.30291277, 0.98852909, 0.7303831 ]])
>>> P
array([[1., 0., 0., 0.],
       [0., 0., 0., 1.],
       [0., 1., 0., 0.],
       [0., 0., 1., 0.]])
>>> L
array([[ 1.        ,  0.        ,  0.        ,  0.        ],
       [ 0.09678623,  1.        ,  0.        ,  0.        ],
       [ 0.73027189, -0.29679299,  1.        ,  0.        ],
       [ 0.39780295, -0.09956144, -0.8465861 ,  1.        ]])
>>> U
array([[0.48657354, 0.72177328, 0.89725033, 0.10555858],
       [0.        , 0.75533446, 0.21016373, 0.84068247],
       [0.        , 0.        , 0.39566752, 0.9028053 ],
       [0.        , 0.        , 0.        , 1.00909401]])
```

Stability of Gaussian Elimination

- Numerical stability of Gaussian Elimination has been an important research topic since the 1940s
- Major figure in this field: James H. Wilkinson (England, 1919–1986)
- Showed that for $Ax = b$ with $A \in \mathbb{R}^{n \times n}$:
  - Gaussian elimination without partial pivoting is numerically unstable
    (as we've already seen)
  - Gaussian elimination with partial pivoting satisfies

$$\frac{\|r\|}{\|A\|\|x\|} \leq 2^{n-1} n^2 \epsilon_{\text{mach}}$$

Stability of Gaussian Elimination

- That is, pathological cases exist where the relative residual $\frac{\|r\|}{\|A\|\|x\|}$ grows exponentially with n due to rounding error
- Worst case behavior of Gaussian Elimination with partial pivoting is explosive instability but such pathological cases are extremely rare!
- In over 50 years of Scientific Computation, instability has only been encountered due to deliberate construction of pathological cases
- In practice, Gaussian elimination is stable in the sense that it produces a small relative residual

Stability of Gaussian Elimination

- In practice, we typically obtain

$$\frac{\|r\|}{\|A\|\|x\|} \lesssim n\epsilon_{\text{mach}}$$

  i.e. grows only linearly with n, and is scaled by ϵ_{mach}
- Combining this result with our inequality $(*)$:

$$\frac{\|\Delta x\|}{\|x\|} \leq \kappa(A)\frac{\|r\|}{\|A\|\|x\|}$$

  implies that in practice Gaussian elimination gives small error for well-conditioned problems!

Cholesky Factorization

Cholesky Factorization

- Suppose that matrix $A \in \mathbb{R}^{n \times n}$ is
  - symmetric: $A^T = A$
  - positive definite: for any $x \neq 0$, $x^T A x > 0$
- Then the matrix can be represented as

$$A = LL^T$$

  known as Cholesky factorization,
  where $L \in \mathbb{R}^{n \times n}$ is a lower triangular matrix
- In general, any matrix of the form BB^T
  is symmetric and positive definite for any $B \in \mathbb{R}^{n \times n}$

Cholesky Factorization

- Matrix L is found directly from equation

$$A = LL^T$$

- Consider the 3×3 case

$$\begin{bmatrix} a_{11} & * & * \\ a_{21} & a_{22} & * \\ a_{31} & a_{32} & a_{33} \end{bmatrix} = \begin{bmatrix} \ell_{11}^2 & * & * \\ \ell_{11}\ell_{21} & \ell_{21}^2 + \ell_{22}^2 & * \\ \ell_{11}\ell_{31} & \ell_{21}\ell_{31} + \ell_{22}\ell_{32} & \ell_{31}^2 + \ell_{32}^2 + \ell_{33}^2 \end{bmatrix}$$

- Equate components starting with the first column

$$\begin{array}{l|l|l} \ell_{11} = \sqrt{a_{11}} & & \\ \ell_{21} = a_{21}/\ell_{11} & \ell_{22} = \sqrt{a_{22} - \ell_{21}^2} & \\ \ell_{31} = a_{31}/\ell_{11} & \ell_{32} = (a_{32} - \ell_{21}\ell_{31})/\ell_{22} & \ell_{33} = \sqrt{a_{33} - \ell_{31}^2 - \ell_{32}^2} \end{array}$$

Cholesky Factorization

- The same approach is generalized to the $n \times n$ case

```
L = 0
ℓ_ij = a_ij for i = 1, ..., n, j = 1, ..., i
for j = 1 : n do
    ℓ_jj = √ℓ_jj
    for i = j + 1 : n do
        ℓ_ij = ℓ_ij / ℓ_jj
    end for
    for k = j + 1 : n do
        for i = k : n do
            ℓ_ik = ℓ_ik − ℓ_ij ℓ_kj
        end for
    end for
end for
```

Cholesky Factorization

- Notes on Cholesky factorization
  - Cholesky factorization is numerically stable and does not require pivoting
  - Operation count: $\sim \frac{1}{3}n^3$ operations in total, i.e. about half as many as Gaussian elimination
  - Only need to store L, so uses less memory than LU. Can be done in-place, overwriting matrix A
- See [examples/unit2/cholesky.py]

Performance Metrics

Performance Metrics

- There are various metrics for software performance
  - performance (FLOP/s): floating point operations per second
  - time to solution
  - scaling efficiency (for parallel computing)
- High Performance Computing studies and develops efficient implementations of numerical algorithms
- Naive Python implementations (e.g. using for-loops) are typically slow
- Modules such as NumPy rely on faster implementations (e.g. written in C)
- Example of performance measurements for Cholesky factorization
  - Python [examples/unit2/cholesky_time.py]
  - C++ [examples/unit2/cholesky_time.cpp]

Sparse Matrices

- In applications, we often encounter sparse matrices
- Common example: discretizations of partial differential equations
- The term sparse matrix typically means that the number of non-zero elements is comparable to the number of rows or columns (e.g. $n \times n$ matrix with $\mathcal{O}(n)$ non-zeros)
- It is advantageous to store and operate only on non-zero elements
- Positions of non-zero elements of a sparse matrix form its sparsity pattern
- Matrices that are not sparse are called dense matrices

Sparse Matrices

- Dense matrices are typically stored as two-dimensional arrays
- Sparse matrices benefit from special data structures and algorithms for computational efficiency
- Example from Unit 1 (constructing a spline)
  - a tridiagonal matrix is stored as three one-dimensional arrays
  - the linear system is solved using the TDMA algorithm
- Standard algorithms (e.g. LU or Cholesky factorization) can be directly applied to sparse matrices. However, new non-zero elements will appear
- These new non-zero elements are called the fill-in.
  Fill-in can be reduced by permuting rows and columns of the matrix
- `scipy.sparse` implements sparse linear algebra

Sparse Matrices: Data Structures

- Coordinate format (COO):
  Arrays: `data`, `row`, `col`
  Element `data[k]` is in row `row[k]` and column `col[k]`
- Compressed Sparse Row (CSR):
  Arrays: `data`, `indices`, `indptr`
  Row `i` contains elements `data[indptr[i]:indptr[i+1]]`
  in columns `indices[indptr[i]:indptr[i+1]]`
- Compressed Sparse Column (CSC):
  Arrays: `data`, `indices`, `indptr`
  Column `j` contains elements `data[indptr[j]:indptr[j+1]]`
  in rows `indices[indptr[j]:indptr[j+1]]`

Example: Sparse Matrix

$$\begin{bmatrix} a & b & b & b & b \\ 0 & c & 0 & 0 & 0 \\ 0 & 0 & c & 0 & 0 \\ 0 & 0 & 0 & c & 0 \end{bmatrix}$$

- Coordinate format (COO):
  $\texttt{data} = (a, b, b, b, b, c, c, c)$
  $\texttt{row} = (0, 0, 0, 0, 0, 1, 2, 3)$
  $\texttt{col} = (0, 1, 2, 3, 4, 1, 2, 3)$
  (assume zero-based indexing)
- See [examples/unit2/sparse.py]

- Compressed Sparse Row (CSR):
  $\texttt{data} = (a, b, b, b, b, c, c, c)$
  $\texttt{indices} = (0, 1, 2, 3, 4, 1, 2, 3)$
  $\texttt{indptr} = (0, 5, 6, 7, 8)$
- Compressed Sparse Column (CSC)
  $\texttt{data} = (a, b, c, b, c, b, c, b)$
  $\texttt{indices} = (0, 0, 1, 0, 2, 0, 3, 0)$
  $\texttt{indptr} = (0, 1, 3, 5, 7, 8)$

QR Factorization

- A square matrix $Q \in \mathbb{R}^{n \times n}$ is called orthogonal if its columns and rows are orthonormal vectors
- Equivalently, $Q^T Q = Q Q^T = \mathrm{I}$
- Orthogonal matrices preserve the Euclidean norm of a vector

$$\|Qv\|_2^2 = v^T Q^T Q v = v^T v = \|v\|_2^2$$

- Geometrically, orthogonal matrices correspond to reflection or rotation
- Orthogonal matrices are very important in scientific computing, norm-preservation implies no amplification of numerical error!

QR Factorization

- The full QR factorization of matrix $A \in \mathbb{R}^{m \times n}$, $m \geq n$ has the form

$$A = QR$$

  where
  - $Q \in \mathbb{R}^{m \times m}$ is orthogonal
  - $R = \begin{bmatrix} \hat{R} \\ 0 \end{bmatrix} \in \mathbb{R}^{m \times n}$
  - $\hat{R} \in \mathbb{R}^{n \times n}$ is upper-triangular
- QR is used for solving overdetermined linear least-squares problems
- QR can be used for solving square systems, but requires twice as many operations as Gaussian elimination

QR Factorization

- Consider the 2-norm of the least-squares residual

$$\|r(x)\|_2^2 = \|b - Ax\|_2^2 = \left\| b - Q \begin{bmatrix} \hat{R} \\ 0 \end{bmatrix} x \right\|_2^2 =$$

$$= \left\| Q^T \left(b - Q \begin{bmatrix} \hat{R} \\ 0 \end{bmatrix} x \right) \right\|_2^2 = \left\| Q^T b - \begin{bmatrix} \hat{R} \\ 0 \end{bmatrix} x \right\|_2^2$$

- Denote $\begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = Q^T b$ with $c_1 \in \mathbb{R}^n, c_2 \in \mathbb{R}^{m-n}$, so that

$$\|r(x)\|_2^2 = \left\| \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} - \begin{bmatrix} \hat{R} \\ 0 \end{bmatrix} x \right\|_2^2 = \left\| \begin{bmatrix} c_1 - \hat{R}x \\ c_2 \end{bmatrix} \right\|_2^2 = \|c_1 - \hat{R}x\|_2^2 + \|c_2\|_2^2$$

QR Factorization

- Question: How do we choose x to minimize $\|r(x)\|_2$?

$$\|r(x)\|_2^2 = \|c_1 - \hat{R}x\|_2^2 + \|c_2\|_2^2$$

where $A = Q\begin{bmatrix}\hat{R}\\0\end{bmatrix}$ and $\begin{bmatrix}c_1\\c_2\end{bmatrix} = Q^T b$

- Answer: Only the first term depends on x. Try setting the first term to zero, i.e. solve the $n \times n$ triangular system

$$\hat{R}x = c_1$$

- This is what `numpy.linalg.lstsq()` does
- Also, this implies that $\min_{x\in\mathbb{R}^n} \|r(x)\|_2 = \|c_2\|_2$

QR Factorization

- Recall that solving linear least-squares via the normal equations requires solving a system with the matrix $A^T A$
- But using the normal equations directly is problematic since

$$\kappa(A^T A) = \kappa(A)^2$$

  (with $\kappa(A)$ for rectangular A defined using SVD, to be covered soon)
- The QR approach avoids this condition-squaring effect and is much more numerically stable!

QR Factorization

- How do we compute the QR factorization?
- There are three main methods
  - Gram–Schmidt orthogonalization
  - Householder triangularization
  - Givens rotations

Gram–Schmidt Orthogonalization

- Suppose $A \in \mathbb{R}^{m \times n}$, $m \geq n$
- One way to picture the QR factorization is to construct a sequence of orthonormal vectors $q_1, q_2, \ldots$ such that

$$\text{span}\{q_1, q_2, \ldots, q_j\} = \text{span}\{a_{(:,1)}, a_{(:,2)}, \ldots, a_{(:,j)}\}, \quad j = 1, \ldots, n$$

- We seek coefficients r_{ij} such that

$$\begin{aligned} a_{(:,1)} &= r_{11} q_1 \\ a_{(:,2)} &= r_{12} q_1 + r_{22} q_2 \\ &\ldots \\ a_{(:,n)} &= r_{1n} q_1 + r_{2n} q_2 + \ldots + r_{nn} q_n \end{aligned}$$

- This can be done via the Gram–Schmidt process

Gram–Schmidt Orthogonalization

- In matrix form we have:

$$\left[\begin{array}{c|c|c|c} a_{(:,1)} & a_{(:,2)} & \cdots & a_{(:,n)} \end{array}\right] = \left[\begin{array}{c|c|c|c} q_1 & q_2 & \cdots & q_n \end{array}\right] \begin{bmatrix} r_{11} & r_{12} & \cdots & r_{1n} \\ & r_{22} & & r_{2n} \\ & & \ddots & \vdots \\ & & & r_{nn} \end{bmatrix}$$

- This gives $A = \hat{Q}\hat{R}$ for $\hat{Q} \in \mathbb{R}^{m\times n}$, $\hat{R} \in \mathbb{R}^{n\times n}$
- This is called the reduced QR factorization of A,
  which is different from the full QR factorization: Q is non-square
- Note that for $m > n$, $\hat{Q}^T\hat{Q} = \mathrm{I}$, but $\hat{Q}\hat{Q}^T \neq \mathrm{I}$

Full vs Reduced QR Factorization

- To obtain the full QR factorization defined earlier

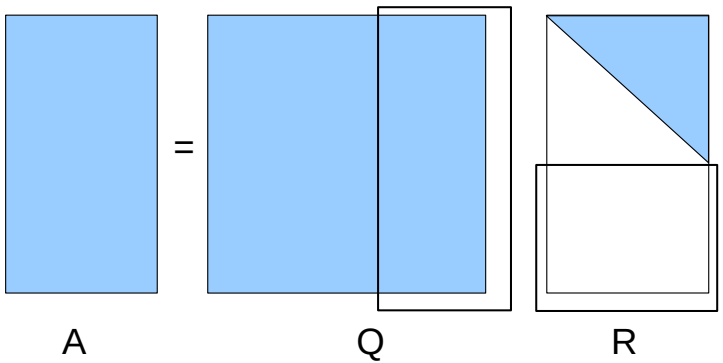
$$A = QR$$

  - append $\hat{Q}$ by $m - n$ arbitrary columns
    that are linearly independent with columns of $\hat{Q}$
  - apply the Gram–Schmidt process to obtain an orthogonal $Q \in \mathbb{R}^{m \times m}$

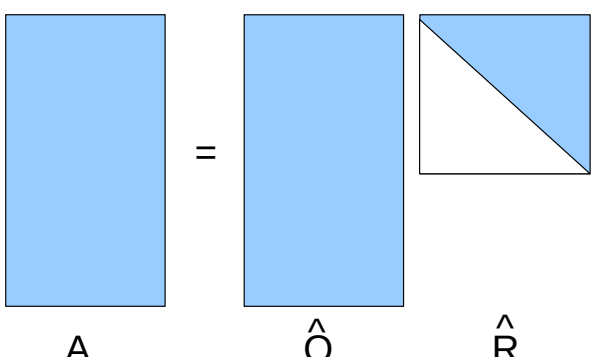
- We also need to append $\hat{R}$ with zero rows to obtain $R = \begin{bmatrix} \hat{R} \\ 0 \end{bmatrix} \in \mathbb{R}^{m \times n}$

  so that the new arbitrary columns in Q do not affect the product

Full vs Reduced QR Factorization



Full QR



Reduced QR

Full vs Reduced QR Factorization

- Exercise: Show that the linear least-squares solution is given by

$$\hat{R}x = \hat{Q}^T b$$

  by plugging $A = \hat{Q}\hat{R}$ into the normal equations
- This is equivalent to the least-squares result
  we showed earlier using the full QR factorization, since $c_1 = \hat{Q}^T b$

Full vs. Reduced QR Factorization

- By default, `numpy.linalg.qr()` does reduced QR factorization

```
>>> import numpy as np
>>> np.random.seed(2022)
>>> a = np.random.random((4,2))
>>> a
array([[0.00935861, 0.49905781],
       [0.11338369, 0.04997402],
       [0.68540759, 0.48698807],
       [0.89765723, 0.64745207]])
>>> (q, r) = np.linalg.qr(a)
>>> q
array([[-0.00824455,  0.99789386],
       [-0.09988626, -0.06374317],
       [-0.60381526, -0.01057732],
       [-0.79079826,  0.00572413]])
>>> r
array([[-1.13512797, -0.81516102],
       [ 0.        ,  0.4933763 ]])
```

- Supplying `mode="complete"` option gives complete QR factorization

```
>>> import numpy as np
>>> np.random.seed(2022)
>>> a = np.random.random((4,2))
>>> a
array([[0.00935861, 0.49905781],
       [0.11338369, 0.04997402],
       [0.68540759, 0.48698807],
       [0.89765723, 0.64745207]])
>>> (q, r) = np.linalg.qr(a, mode="complete")
>>> q
array([[-0.00824455,  0.99789386, -0.02953283, -0.0571
       [-0.09988626, -0.06374317, -0.61111959, -0.7826
       [-0.60381526, -0.01057732,  0.66414863, -0.4406
       [-0.79079826,  0.00572413, -0.42961291,  0.4359
>>> r
array([[-1.13512797, -0.81516102],
       [ 0.        ,  0.4933763 ],
       [ 0.        ,  0.        ],
       [ 0.        ,  0.        ]])
```

Gram–Schmidt Orthogonalization

- Returning to the Gram–Schmidt process, how do we compute the q_i, $i = 1, \ldots, n$?
- In the j-th step, find a unit vector $q_j \in \text{span}\{a_{(:,1)}, a_{(:,2)}, \ldots, a_{(:,j)}\}$ that is orthogonal to $\text{span}\{q_1, q_n, \ldots, q_{j-1}\}$
- We set

$$v_j = a_{(:,j)} - \sum_{i=1}^{j-1} (q_i^T a_{(:,j)}) q_i$$

  and then set $q_j = v_j / \|v_j\|_2$
- Exercise: Verify that q_j satisfies the requirements
- We can now determine the required values of r_{ij}

Gram–Schmidt Orthogonalization

- From the equations $A = \hat{Q}\hat{R}$, for $j = 1, \ldots, n$

$$q_j = \frac{a_{(:,j)} - \sum_{i=1}^{j-1} r_{ij} q_i}{r_{jj}}$$

- From the Gram–Schmidt process, for $j = 1, \ldots, n$

$$q_j = \frac{a_{(:,j)} - \sum_{i=1}^{j-1} (q_i^T a_{(:,j)}) q_i}{\|a_{(:,j)} - \sum_{i=1}^{j-1} (q_i^T a_{(:,j)}) q_i\|_2}$$

- Both expressions have the same structure, by matching the terms

$$\begin{aligned} r_{ij} &= q_i^T a_{(:,j)} \qquad (i \neq j) \\ |r_{jj}| &= \|a_{(:,j)} - \textstyle\sum_{i=1}^{j-1} r_{ij} q_i\|_2 \end{aligned}$$

- The sign of r_{jj} is not determined uniquely, so we can choose $r_{jj} > 0$

Classical Gram–Schmidt Process

- The resulting algorithm is referred to as the classical Gram–Schmidt (CGS) method

```
for j = 1 : n do
    v_j = a_(:,j)
    for i = 1 : j − 1 do
        r_ij = q_i^T a_(:,j)
        v_j = v_j − r_ij q_i
    end for
    r_jj = ||v_j||_2
    q_j = v_j / r_jj
end for
```

Gram–Schmidt Orthogonalization

- The only way the Gram–Schmidt process can fail is if $|r_{jj}| = \|v_j\|_2 = 0$ for some j
- This can only happen if $a_{(:,j)} = \sum_{i=1}^{j-1} r_{ij} q_i$ for some j, i.e. if $a_{(:,j)} \in \operatorname{span}\{q_1, q_n, \ldots, q_{j-1}\} = \operatorname{span}\{a_{(:,1)}, a_{(:,2)}, \ldots, a_{(:,j-1)}\}$
- This means that columns of A are linearly dependent
- Therefore, Gram–Schmidt fails $\implies$ columns of A linearly dependent

Gram–Schmidt Orthogonalization

- Therefore, if columns of A are linearly independent, then the Gram–Schmidt succeeds
- The only non-uniqueness in the Gram–Schmidt process was in the sign of r_{ii}, therefore $\hat{Q}\hat{R}$ is unique under the requirement that all $r_{ii} > 0$
- This proves the following
  Theorem: Every $A \in \mathbb{R}^{m \times n} (m \geq n)$ of full rank has a unique reduced QR factorization $A = \hat{Q}\hat{R}$ with $r_{ii} > 0$

Gram–Schmidt Orthogonalization

- Theorem: Every $A \in \mathbb{R}^{m \times n} (m \geq n)$ has a full QR factorization
- Case 1: A has full rank
  - we compute the reduced QR factorization from above
  - to make Q square we pad $\hat{Q}$ with $m - n$ arbitrary orthonormal columns
  - we also pad $\hat{R}$ with $m - n$ zero rows to get R
- Case 2: A does not have full rank
  - at some point in computing the reduced QR factorization, we encounter $\|v_j\|_2 = 0$
  - at this point we pick an arbitrary unit q_j orthogonal to $\text{span}\{q_1, q_2, \ldots, q_{j-1}\}$ and then proceed as in Case 1

Modified Gram–Schmidt Process

- The classical Gram–Schmidt process is numerically unstable!
  (sensitive to rounding error, orthogonality of the q_j degrades)
- The algorithm can be reformulated to give
  the modified Gram–Schmidt process,
  which is numerically more robust
- Key idea: when each new q_j is computed,
  orthogonalize each remaining column of A against it

Modified Gram–Schmidt Process

- Applying this idea results in the modified Gram–Schmidt (MGS) method

```
for i = 1 : n do
    v_i = a_(:,i)
end for
for i = 1 : n do
    r_ii = ||v_i||_2
    q_i = v_i / r_ii
    for j = i + 1 : n do
        r_ij = q_i^T v_j
        v_j = v_j − r_ij q_i
    end for
end for
```

Modified Gram–Schmidt Process

- Key difference between MGS and CGS
  - In CGS we compute orthogonalization coefficients r_{ij} using the original column $a_{(:,j)}$
  - In MGS we remove components of $a_{(:,j)}$ in $\text{span}\{q_1, q_2, \ldots, q_{i-1}\}$ before computing r_{ij}
- This makes no difference mathematically:
  In exact arithmetic components in $\text{span}\{q_1, q_2, \ldots, q_{i-1}\}$ are annihilated by q_i^T
- But in practice it reduces degradation of orthogonality of the q_j and improves the numerical stability of MGS over CGS

Operation Count

- MGS is dominated by the innermost loop (lines 8 and 9):

$$r_{ij} = q_i^T v_j$$
$$v_j = v_j - r_{ij} q_i$$

- The first requires m multiplications, $m-1$ additions; the second requires m multiplications, m subtractions
- Therefore, each innermost iteration takes $\sim 4m$ operations
- The rotal number of operations is asymptotically

$$\sum_{i=1}^{n} \sum_{j=i+1}^{n} 4m \sim 4m \sum_{i=1}^{n} i \sim 2mn^2$$

Alternative QR Factorization Methods

- The QR factorization can also be computed using
  - Householder triangularization
  - Givens rotations
- Both methods apply a sequence of orthogonal matrices

$$Q_1, Q_2, Q_3, \ldots$$

that successively remove terms below the diagonal (similar to the LU factorization)

Householder Triangularization

Householder Triangularization

- We will now discuss the Householder triangularization which is more numerically stable and more efficient than Gram–Schmidt
- Unlike Gram–Schmidt, it will not guarantee that the orthonormal basis at each step will span the same subspaces as columns of A

$$\operatorname{span}\{a_{(:,1)}\}, \quad \operatorname{span}\{a_{(:,1)}, a_{(:,2)}\}, \quad \ldots$$

  which may be important for some applications
- Method used by `scipy.linalg.qr()` calling `dgeqrf()` from LAPACK
- Introduced by Alston Householder (1904–1993, USA)

Householder Triangularization

- Idea: Apply a succession of orthogonal matrices $Q_k \in \mathbb{R}^{m \times m}$ to A to compute an upper triangular matrix R

$$R = Q_n \cdots Q_2 Q_1 A$$

- That will result in the full QR factorization

$$A = QR$$

since $Q = Q_1^T Q_2^T \ldots Q_n^T$ is a square matrix

Householder Triangularization

- In 1958, Householder proposed a way to choose Q_k to introduce zeros below the diagonal in column k while preserving the previous columns

$$\underbrace{\begin{bmatrix} * & * & * \\ * & * & * \\ * & * & * \\ * & * & * \\ * & * & * \end{bmatrix}}_{A} \xrightarrow{Q_1} \underbrace{\begin{bmatrix} * & * & * \\ 0 & * & * \\ 0 & * & * \\ 0 & * & * \\ 0 & * & * \end{bmatrix}}_{Q_1 A} \xrightarrow{Q_2} \underbrace{\begin{bmatrix} * & * & * \\ 0 & * & * \\ 0 & 0 & * \\ 0 & 0 & * \\ 0 & 0 & * \end{bmatrix}}_{Q_2 Q_1 A} \xrightarrow{Q_3} \underbrace{\begin{bmatrix} * & * & * \\ 0 & * & * \\ 0 & 0 & * \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}}_{Q_3 Q_2 Q_1 A}$$

- This is achieved by Householder reflectors

Householder Reflectors

- We choose

$$Q_k = \begin{bmatrix} \mathrm{I}_{k-1} & 0 \\ 0 & F \end{bmatrix}$$

  - $\mathrm{I}_{k-1} \in \mathbb{R}^{(k-1)\times(k-1)}$
  - $F \in \mathbb{R}^{(m-k+1)\times(m-k+1)}$ is a Householder reflector
- The I_{k-1} block ensures the first $k-1$ rows are unchanged
- F is an orthogonal matrix that operates on the bottom $m-k+1$ rows
- If F is orthogonal, then Q_k is orthogonal

Householder Reflectors

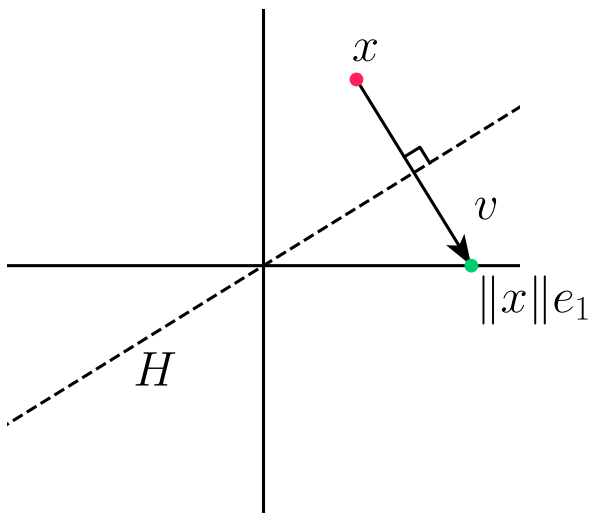
- Let $x \in \mathbb{R}^{m-k+1}$ denote elements $k, \ldots, m$ of the k-th column in the current matrix $Q_{k-1} \ldots Q_1 A$
- We have two requirements for F
  1. F is orthogonal, in particular $\|Fx\|_2 = \|x\|_2$
  2. only the first element of Fx is non-zero
- Therefore, we must have

$$Fx = F \begin{bmatrix} * \\ * \\ \vdots \\ * \end{bmatrix} = \begin{bmatrix} \|x\|_2 \\ 0 \\ \vdots \\ 0 \end{bmatrix} = \|x\|_2 e_1$$

- Question: How can we achieve this?

Householder Reflectors

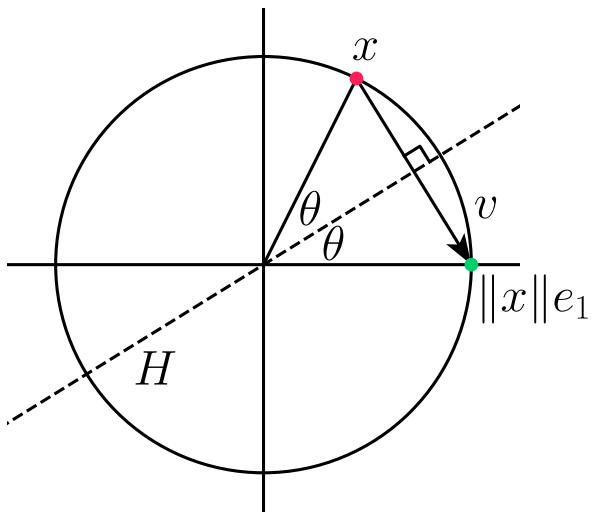
- We can see geometrically that this can be achieved by reflection across a hyperplane H



- Here H is the hyperplane orthogonal to $v = \|x\|e_1 - x$, and the key point is that H passes through the origin 0

Householder Reflectors

- H passes through the origin because x and $\|x\|e_1$
  both belong to the hypersphere with radius $\|x\|_2$ centered at the origin



- Also analytically, since $(x + \|x\|e_1)/2 \in H$,
  we have $0 \in H \iff (\|x\|e_1 - x) \cdot (x + \|x\|e_1) = \|x\|^2 - x \cdot x = 0$

Householder Reflectors

- Next, we need to determine the matrix F which maps x to $\|x\|_2 e_1$
- F is closely related to the orthogonal projection of x onto H, since that projection takes us "half way" from x to $\|x\|_2 e_1$
- Hence we first consider orthogonal projection onto H, and subsequently derive F

Householder Reflectors

- The orthogonal projection of vector a onto vector b is given by

$$\frac{(a \cdot b)}{||b||^2} b$$

since $\left(a - \frac{(a \cdot b)}{||b||^2} b\right) \cdot b = a \cdot b - \frac{(a \cdot b)}{||b||^2} b \cdot b = 0$

- In the matrix form

$$\frac{(a \cdot b)}{||b||^2} b = \frac{1}{b^T b}(a^T b) b = \frac{1}{b^T b} b(b^T a) = \left(\frac{1}{b^T b} b b^T\right) a$$

- Therefore, the matrix $\frac{1}{b^T b} b b^T$ orthogonally projects onto b

Householder Reflectors

- We have that $\frac{1}{v^T v} v v^T$ orthogonally projects onto v
- Then, the following matrix

$$P_H = \mathrm{I} - \frac{v v^T}{v^T v}$$

  orthogonally projects onto H as it satisfies
  - $P_H x \in H$
    since $v^T P_H x = v^T x - v^T \frac{v v^T}{v^T v} x = v^T x - \frac{v^T v}{v^T v} v^T x = 0$
  - $x - P_H x$ is orthogonal to H
    since $x - P_H x = x - x + \frac{v v^T}{v^T v} x = \frac{v^T x}{v^T v} v$ is proportional to v

Householder Reflectors

- But recall that F should reflect across H rather than project onto H

$$P_H = \mathrm{I} - \frac{vv^T}{v^T v}$$

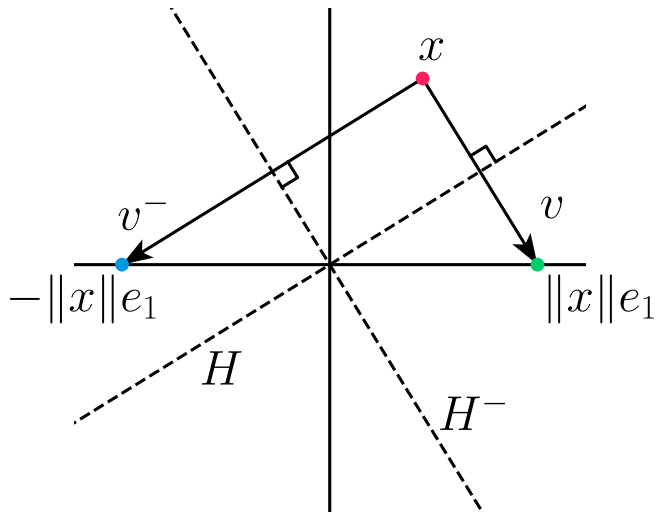
- We obtain F by going "twice as far" in the direction of v compared to P_H

$$F = \mathrm{I} - 2\frac{vv^T}{v^T v}$$

- Exercise: Show that F is an orthogonal matrix, i.e. that $F^T F = \mathrm{I}$

Householder Reflectors

- In fact, there are two Householder reflectors that we can choose from



- Which one is better?

Householder Reflectors

- If x and $\|x\|_2 e_1$ (or x and $-\|x\|_2 e_1$) are close, we could obtain loss of precision due to cancellation when computing $v = \|x\| e_1 - x$ (or $v = -\|x\| e_1 - x$)
- To ensure x and its reflection are well separated we should choose the reflection to be

$$-\operatorname{sign}(x_1)\|x\|_2 e_1$$

- Therefore, we want to have $v = -\operatorname{sign}(x_1)\|x\|_2 e_1 - x$
- Since the sign of v does not affect F, we scale v by -1 to get

$$v = \operatorname{sign}(x_1)\|x\|_2 e_1 + x$$

Householder Reflectors

- Let's compare the two options for v in the potentially problematic case when $x \approx \|x\|_2 e_1$, i.e. when $x_1 \approx \|x\|_2$
  - $v_{\text{bad}} = \|x\|_2 e_1 - x$
  - $v_{\text{good}} = \text{sign}(x_1)\|x\|_2 e_1 + x$
- The corresponding norms are

$$\|v_{\text{bad}}\|_2^2 = \left\| \|x\|_2 e_1 - x \right\|_2^2 \approx 0$$

$$\begin{aligned}
\|v_{\text{good}}\|_2^2 &= \left\| \text{sign}(x_1)\|x\|_2 e_1 + x \right\|_2^2 \\
&= (\text{sign}(x_1)\|x\|_2 + x_1)^2 + \|x_{(2:m-k+1)}\|_2^2 \\
&= (\text{sign}(x_1)\|x\|_2 + \text{sign}(x_1)|x_1|)^2 + \|x_{(2:m-k+1)}\|_2^2 \\
&= (\|x\|_2 + |x_1|)^2 + \|x_{(2:m-k+1)}\|_2^2 \approx (2\|x\|_2)^2
\end{aligned}$$

Householder Reflectors

- Recall that v is computed from two vectors of magnitude $\|x\|_2$
- The argument above shows that with v_{bad} we can get $\|v\|_2 \ll \|x\|_2$ leading to loss of precision due to cancellation
- In contrast, with v_{good} we always have $\|v_{\text{good}}\|_2 \geq \|x\|_2$, which rules out loss of precision due to cancellation

Householder Triangularization

- We can now write out the Householder algorithm

```
1: for k = 1 : n do
2:     x = a_(k:m,k)
3:     v_k = sign(x_1)||x||_2 e_1 + x
4:     v_k = v_k / ||v_k||_2
5:     a_(k:m,k:n) = a_(k:m,k:n) − 2v_k (v_k^T a_(k:m,k:n))
6: end for
```

- It overwrites A with R and stores $v_1, \ldots, v_n$
- Note that we do not divide by $v_k^T v_k$ in line 5 since we normalize v_k in line 4
- Householder algorithm requires $\sim 2mn^2 - \frac{2}{3}n^3$ operations (while Gram–Schmidt requires $2mn^2$)

Householder Triangularization

- Note that we do not explicitly form Q
- We can use the vectors $v_1, \ldots, v_n$ to compute Q in a post-processing step
- Recall that

$$Q_k = \begin{bmatrix} \mathrm{I} & 0 \\ 0 & F \end{bmatrix}$$

  and $Q = (Q_n \cdots Q_2 Q_1)^T = Q_1^T Q_2^T \cdots Q_n^T$
- Also, the Householder reflectors are symmetric (see the definition of F), so $Q = Q_1^T Q_2^T \cdots Q_n^T = Q_1 Q_2 \cdots Q_n$ and
- Note that each Q_k is involutory (i.e. $Q_k^{-1} = Q_k$) but in general this does not hold for the product ($Q^{-1} \neq Q$)

Householder Triangularization

- For any y, we can evaluate $Qy = Q_1 Q_2 \cdots Q_n y$ using the v_k

```
for k = n : -1 : 1 do
    y_(k:m) = y_(k:m) - 2 v_k (v_k^T y_(k:m))
end for
```

- Question: How can we use this to form the matrix Q?

Householder Triangularization

- Answer: Compute Q from Qe_i, $i = 1, \ldots, m$
  since Q consists of columns Qe_i
- Similarly, compute the reduced $\hat{Q}$ from Qe_i, $i = 1, \ldots, n$
- However, often not necessary to form Q or $\hat{Q}$ explicitly,
  e.g. to solve the least-squares problem $Ax \simeq b$,
  we only need the product $Q^T b$ and the matrix R
- Note that the product $Q^T b = Q_n \cdots Q_2 Q_1 b$ can be evaluated as

```
for k = 1 : n do
    b_(k:m) = b_(k:m) - 2 v_k (v_k^T b_(k:m))
end for
```

Givens Rotations

Givens Rotations

- Another method of QR-factorization is based on Givens rotation matrix

$$G(i,j,\theta) = \begin{pmatrix} 1 & \dots & 0 & \dots & 0 & \dots & 0 \\ \vdots & \ddots & \vdots & \ddots & \vdots & \ddots & \vdots \\ 0 & \dots & c & \dots & -s & \dots & 0 \\ \vdots & \ddots & \vdots & \ddots & \vdots & \ddots & \vdots \\ 0 & \dots & s & \dots & c & \dots & 0 \\ \vdots & \ddots & \vdots & \ddots & \vdots & \ddots & \vdots \\ 0 & \dots & 0 & \dots & 0 & \dots & 1 \end{pmatrix}$$

which is defined for $i < j$ and $\theta \in \mathbb{R}$ as an $m \times m$ matrix with elements

$$g_{ii} = c, \qquad g_{jj} = c, \qquad g_{ij} = -s, \qquad g_{ji} = s$$
$$g_{kk} = 1 \quad \text{for } k \neq i, j, \qquad g_{kl} = 0 \quad \text{otherwise}$$

where $c = \cos\theta$ and $s = \sin\theta$

Givens Rotations

- A Givens rotation matrix applies a rotation within the space spanned by the i-th and j-th coordinates
- Named after James W. Givens, Jr. (1910–1993, USA)

Effect of a Givens rotation

- Consider a rectangular matrix $A \in \mathbb{R}^{m \times n}$ where $m \geq n$
- Suppose that a_1 and a_2 are in the i-th and j-th positions in a particular column of A. Assume that $a_1^2 + a_2^2 \neq 0$
- Restricting to just i-th and j-th dimensions, a Givens rotation $G(i, j, \theta)$ for a particular angle θ can be chosen so that

$$\begin{pmatrix} c & -s \\ s & c \end{pmatrix} \begin{pmatrix} a_1 \\ a_2 \end{pmatrix} = \begin{pmatrix} \alpha \\ 0 \end{pmatrix}$$

where α is non-zero, and the j-th component is eliminated

Stable computation

- Since the length is preserved, $\alpha = \sqrt{a_1^2 + a_2^2}$
- We could compute

$$c = \frac{a_1}{\sqrt{a_1^2 + a_2^2}}, \qquad s = \frac{-a_2}{\sqrt{a_1^2 + a_2^2}}$$

  but this is susceptible to underflow/overflow if α is very small
- A better procedure is
  - if $|a_1| > |a_2|$, set $t = \tan\theta = a_2/a_1$ and then $c = \frac{1}{\sqrt{1+t^2}}, s = -ct$
  - if $|a_2| \geq |a_1|$, set $t = \cot\theta = a_1/a_2$ and then $s = \frac{-1}{\sqrt{1+t^2}}, c = -st$

Givens rotation algorithm

- The following algorithm performs the full QR-factorization of a matrix $A \in \mathbb{R}^{m \times n}$ with $m \geq n$ using Givens rotations

```
R = A, Q = I
for k = 1 : n do
    for j = m : k + 1 do
        Construct G = G(j − 1, j, θ) to eliminate a_jk
        R = GR
        Q = QG^T
    end for
end for
```

Advantages of Givens Rotations

- In general, for dense matrices, Givens rotations are not as efficient as the other two approaches (Gram–Schmidt and Householder)
- However, they are advantageous for sparse matrices, since non-zero elements can be eliminated one-by-one without affecting other rows

Advantages of Givens Rotations

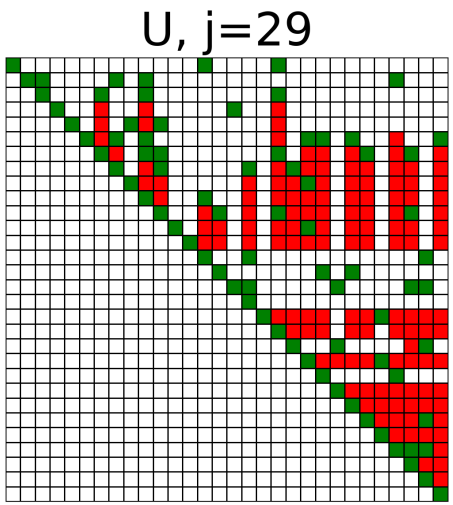
- Also, Givens rotations of different rows can be done concurrently
- Consider the 6×6 matrix

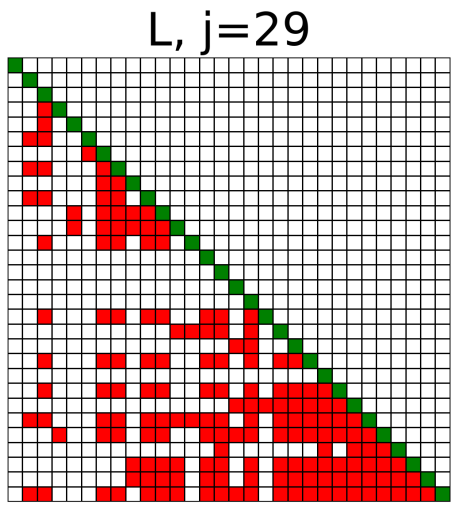
$$\begin{bmatrix} * & * & * & * & * & * \\ 5 & * & * & * & * & * \\ 4 & 6 & * & * & * & * \\ \boxed{3} & 5 & 7 & * & * & * \\ 2 & 4 & 6 & 8 & * & * \\ 1 & \boxed{3} & 5 & 7 & 9 & * \end{bmatrix}$$

- Each number denotes the step when that element can be eliminated
- For example, on step 3, elements $(4, 1)$ and $(6, 2)$ can be eliminated concurrently using $G(3, 4, \cdot)$ and $G(5, 6, \cdot)$ since they operate on different rows

Example: Sparsity Patterns

- Positions of non-zero elements of a sparse matrix form its sparsity pattern
- Transformations of the matrix may introduce new non-zero elements
- These new non-zero elements are called the fill-in
- See [examples/unit2/sparse_pattern.py]

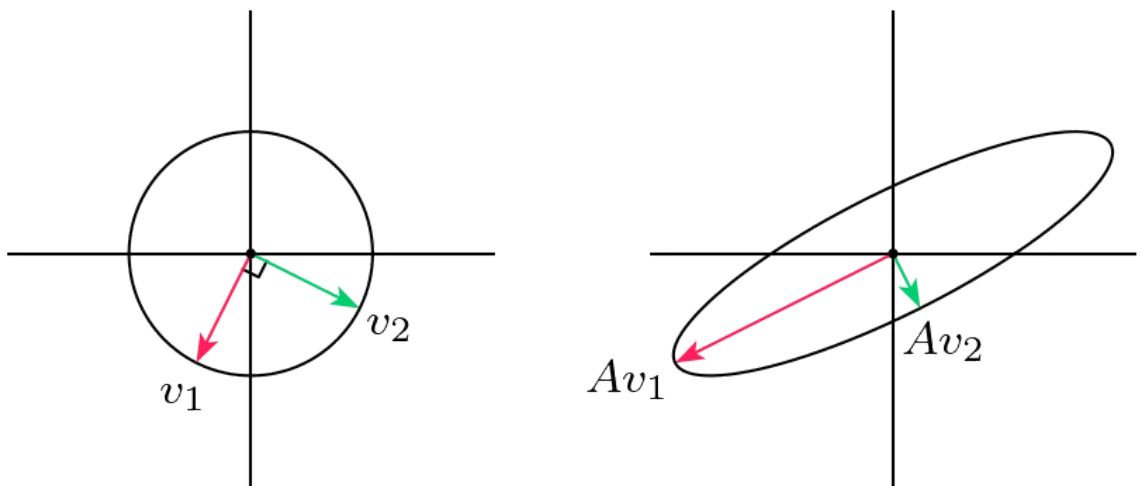




Singular Value Decomposition

Singular Value Decomposition

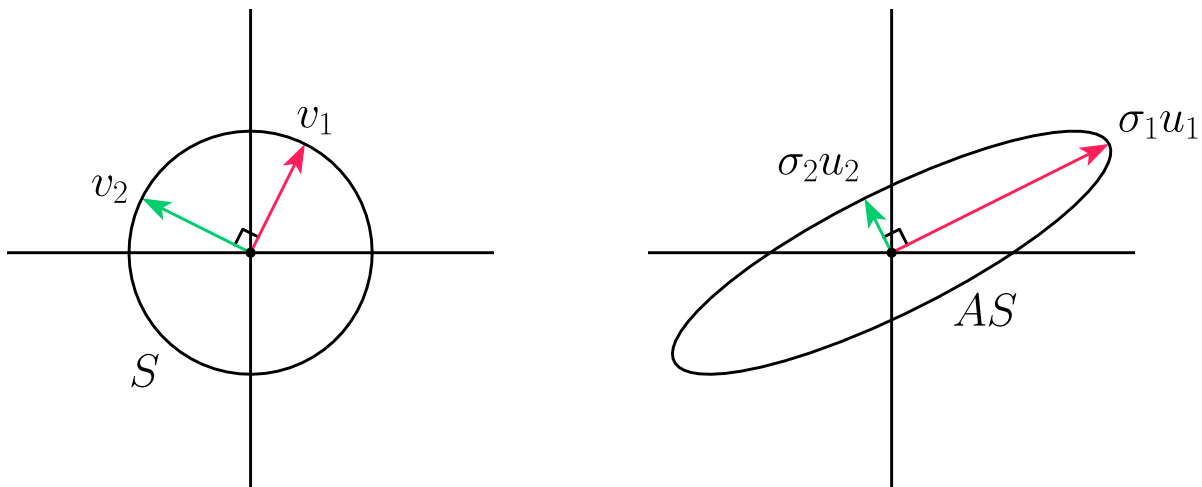
- How does a matrix deform the space?
- Example of $A = \begin{bmatrix} 1 & 1.5 \\ 0 & 1 \end{bmatrix}$ mapping the unit circle to an ellipse



- In general, a matrix does not preserve orthogonality and length

Singular Value Decomposition

- However, orthogonal v_1 and v_2 can be chosen such that $Av_1 = \sigma_1 u_1$ and $Av_2 = \sigma_2 u_2$ are orthogonal



where $\sigma_1 \geq \sigma_2 \geq 0$ and $\|u_1\| = \|u_2\| = 1$

Singular Value Decomposition

- To obtain a Singular Value Decomposition (SVD) of a matrix $A \in \mathbb{R}^{m \times n}$, we are looking for orthonormal vectors v_i such that

$$Av_i = \sigma_i u_i, \quad i = 1, \ldots, n$$

  where vectors u_i are also orthonormal and $\sigma_i \in \mathbb{R},\ \sigma_i \geq 0$

- In the matrix form, we get

$$AV = \hat{U}\hat{\Sigma}$$

$$\begin{bmatrix} & & \\ & A & \\ & & \end{bmatrix} \left[\begin{array}{c|c|c} v_1 & \cdots & v_n \end{array}\right] = \left[\begin{array}{c|c|c} u_1 & \cdots & u_n \end{array}\right] \begin{bmatrix} \sigma_1 & & \\ & \ddots & \\ & & \sigma_n \end{bmatrix}$$

Singular Value Decomposition

- Matrices in $AV = \hat{U}\hat{\Sigma}$ are
  - $A \in \mathbb{R}^{m\times n}$ is a general matrix
  - $V \in \mathbb{R}^{n\times n}$ with orthonormal columns
  - $\hat{\Sigma} \in \mathbb{R}^{n\times n}$ is diagonal with non-negative, real entries
  - $\hat{U} \in \mathbb{R}^{m\times n}$ with orthonormal columns
- Therefore V is an orthogonal matrix ($V^T V = VV^T = \mathrm{I}$) and we have the following decomposition called the reduced SVD

$$A = \hat{U}\hat{\Sigma}V^T$$

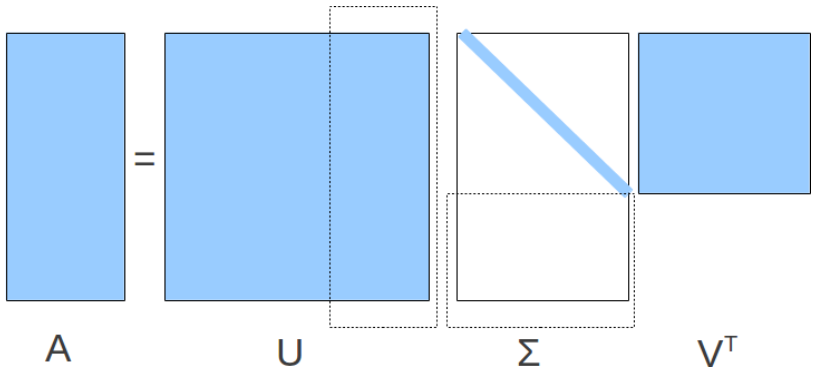
  - $\sigma_1, \sigma_2, \ldots, \sigma_n \geq 0$ are singular values (typically $\sigma_1 \geq \sigma_2 \geq \ldots$)
  - $u_1, u_2, \ldots, u_n$ are left singular vectors (columns of $\hat{U}$)
  - $v_1, v_2, \ldots, v_n$ are right singular vectors (rows of V^T)

Singular Value Decomposition

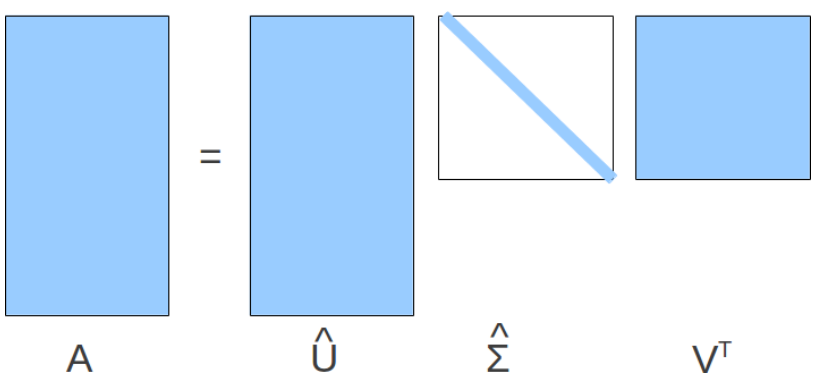
- Just as with QR factorization, we can pad the columns of $\hat{U}$ with $m - n$ arbitrary orthonormal vectors to obtain an orthogonal $U \in \mathbb{R}^{m \times m}$
- We then need to "silence" these arbitrary columns by adding rows of zeros to $\hat{\Sigma} \in \mathbb{R}^{n \times n}$ to obtain $\Sigma \in \mathbb{R}^{m \times n}$
- This gives the full SVD for $A \in \mathbb{R}^{m \times n}$

$$A = U \Sigma V^T$$

Full vs Reduced SVD



Full SVD



Reduced SVD

Singular Value Decomposition

- Theorem: Every matrix $A \in \mathbb{R}^{m \times n}$ has a full singular value decomposition. Furthermore:
  - singular values σ_i are uniquely determined
  - if A is square and σ_j are distinct, then u_i and v_i are uniquely determined up to sign
- Proof is outside of the scope of the course

Singular Value Decomposition

- This theorem justifies the statement:
  the image of the unit hypersphere under any $m \times n$ matrix is a hyperellipse
- Consider $A = U\Sigma V^T$ (full SVD) applied to the unit sphere $S \subset \mathbb{R}^n$:
  - the orthogonal map V^T preserves S
  - Σ stretches S into a hyperellipse aligned with the canonical axes e_j
  - U rotates or reflects the hyperellipse without changing its shape

SVD in Python

- `numpy.linalg.svd()` computes the full SVD by default

```
>>> import numpy as np
>>> np.random.seed(2022)
>>> a=np.random.random((4,2))
>>> a
array([[0.00935861, 0.49905781],
       [0.11338369, 0.04997402],
       [0.68540759, 0.48698807],
       [0.89765723, 0.64745207]])
>>> (u, s, v) = np.linalg.svd(a)
>>> u
array([[-0.22570503,  0.97206861, -0.02953283, -0.0571
       [-0.08357767, -0.08399541, -0.61111959, -0.7826
       [-0.58696968, -0.14202585,  0.66414863, -0.4406
       [-0.77300621, -0.16690133, -0.42961291,  0.4359
>>> s
array([1.42929716, 0.39183261])
>>> v
array([[-0.77506396, -0.63188279],
       [-0.63188279,  0.77506396]])
```

- with `full_matrices=0` it computes the reduced SVD

```
>>> import numpy as np
>>> np.random.seed(2022)
>>> a = np.random.random((4,2))
>>> a
array([[0.00935861, 0.49905781],
       [0.11338369, 0.04997402],
       [0.68540759, 0.48698807],
       [0.89765723, 0.64745207]])
>>> (u, s, v) = np.linalg.svd(a, full_matrices=0)
>>> u
array([[-0.22570503,  0.97206861],
       [-0.08357767, -0.08399541],
       [-0.58696968, -0.14202585],
       [-0.77300621, -0.16690133]])
>>> s
array([1.42929716, 0.39183261])
>>> v
array([[-0.77506396, -0.63188279],
       [-0.63188279,  0.77506396]])
```

Matrix Properties via the SVD

- Let r denote the number of nonzero singular values, so that

$$\sigma_1 \geq \sigma_2 > \cdots \geq \sigma_r > 0, \quad \sigma_{r+1} = \ldots = \sigma_n = 0$$

- Property: $r = \text{rank}(A)$
- Proof: In the full SVD $A = U\Sigma V^T$, matrices U and V^T have full rank, so multiplication by them preserves rank, leading to $\text{rank}(A) = \text{rank}(\Sigma) = r$
- Property: $\text{image}(A) = \text{span}\{u_1, \ldots, u_r\}$ and $\text{null}(A) = \text{span}\{v_{r+1}, \ldots, v_n\}$
- Proof: This follows from $A = U\Sigma V^T$ and

$$\text{image}(\Sigma) = \text{span}\{e_1, \ldots, e_r\} \in \mathbb{R}^m$$
$$\text{null}(\Sigma) = \text{span}\{e_{r+1}, \ldots, e_n\} \in \mathbb{R}^n$$

Matrix Properties via the SVD

- Property: $\|A\|_2 = \sigma_1$
- Proof: By definition $\|A\|_2 = \max_{\|v\|_2=1} \|Av\|_2 = \max_{\|v\|_2=1} \|U\Sigma V^T v\|_2$.
  Orthogonal matrices preserve the norm, $\|A\|_2 = \max_{\|v\|_2=1} \|\Sigma v\|_2 = \sigma_1$
- Property: Singular values of A are the square roots
  of the eigenvalues of $A^T A$ or AA^T
- Proof: $A^T A = (U\Sigma V^T)^T (U\Sigma V^T) = V\Sigma U^T U\Sigma V^T = V(\Sigma^T \Sigma)V^T$
  Therefore, $(A^T A)V = V(\Sigma^T \Sigma)$, or $(A^T A)v_{(:,j)} = \sigma_j^2 v_{(:,j)}$
  (Analogous for AA^T)

Matrix Properties via the SVD

- The pseudoinverse A^+ can be defined more generally in terms of the SVD
- Define pseudoinverse of a scalar $\sigma \in \mathbb{R}$ to be
  $\sigma^+ = 1/\sigma$ if $\sigma \neq 0$ and $\sigma^+ = 0$ if $\sigma = 0$
- Define pseudoinverse $\Sigma^+ \in \mathbb{R}^{n \times m}$ of a diagonal matrix $\Sigma \in \mathbb{R}^{m \times n}$
  as its transpose after taking scalar pseudoinverse of each element
- Define pseudoinverse of $A \in \mathbb{R}^{m \times n}$ as

$$A^+ = V \Sigma^+ U^T$$

- Note: A^+ exists for any matrix A, and it covers
  our previous definitions of pseudoinverse

Matrix Properties via the SVD

- We generalize the condition number to rectangular matrices via the definition

$$\kappa(A) = \|A\|\|A^+\|$$

- Property: The 2-norm condition number is given by

$$\kappa(A) = \sigma_{\max}/\sigma_{\min}$$

- Proof: $\|A\|_2 = \sigma_{\max}$ as shown before.
  The largest singular value of A^+ is $1/\sigma_{\min}$ so $\|A^+\|_2 = 1/\sigma_{\min}$

Matrix Properties via the SVD

- These results indicate the importance of the SVD, both theoretical and as a computational tool
- Algorithms for calculating the SVD are outside scope of this course
- SVD requires $\sim 4mn^2 - \frac{4}{3}n^3$ operations
- For more details on algorithms, see Trefethen & Bau, or Golub & van Loan

Low-Rank Approximation via the SVD

- One of the most useful properties of the SVD is that it allows us to obtain an optimal low-rank approximation to A
- We can recast SVD as

$$A = \sum_{j=1}^{r} \sigma_j u_j v_j^T$$

- Follows from writing Σ as a sum of r matrices Σ_j, where $\Sigma_j = \text{diag}(0, \ldots, 0, \sigma_j, 0, \ldots, 0)$
- Each $u_j v_j^T$ is a rank one matrix: each column is a scaled version of u_j

Low-Rank Approximation via the SVD

- Theorem: For any index $\nu = 0, \ldots, r$ the matrix

$$A_\nu = \sum_{j=1}^{\nu} \sigma_j u_j v_j^T$$

satisfies

$$\|A - A_\nu\|_2 = \inf_{B \in \mathbb{R}^{m \times n},\, \mathrm{rank}(B) \leq \nu} \|A - B\|_2 = \sigma_{\nu+1}$$

- That is
  - A_ν is the closest rank ν matrix to A, measured in the 2-norm
  - The error in A_ν is given by the first omitted singular value

Low-Rank Approximation via the SVD

- A similar result holds in the Frobenius norm:

$$\|A - A_\nu\|_F = \inf_{B \in \mathbb{R}^{m \times n},\, \operatorname{rank}(B) \leq \nu} \|A - B\|_F = \sqrt{\sigma_{\nu+1}^2 + \cdots + \sigma_r^2}$$

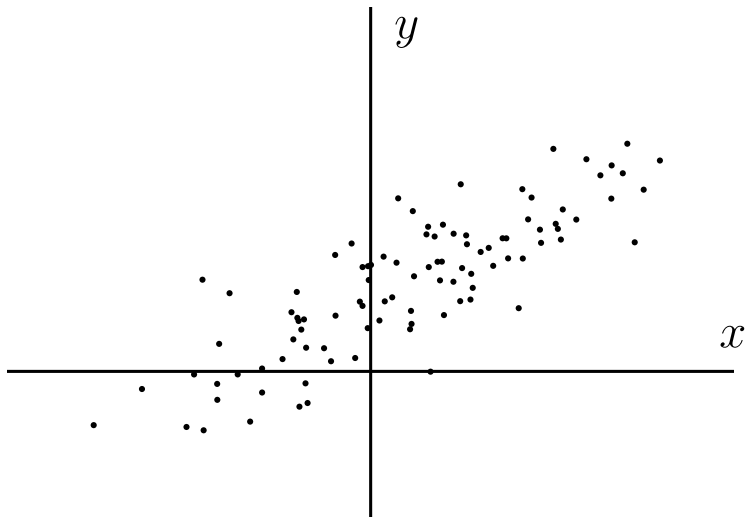
Low-Rank Approximation via the SVD

- These theorems indicate that the SVD is an effective way to compress data encapsulated by a matrix!
- For example, A can represent an image
- If singular values of A decay rapidly,
  we can approximate A with few rank one matrices
- For each rank one matrix $\sigma_j u_j v_j$,
  we only need to store $m + n + 1$ numbers: $\sigma_j,\ u_j,\ v_j$

Principal Component Analysis

Principal Component Analysis

- Consider a dataset of $(x_i, y_i) \in \mathbb{R}^2$ for $i = 1, \ldots, m$



- There is a strong correlation between x and y
- This means that we can describe most of the data with just one feature
- This is done by Principal Component Analysis (PCA)

Principal Component Analysis

- The new axis should maximize variance of the data
- Consider the empirical covariance matrix

$$M = \begin{bmatrix} \mathrm{Var}(x) & \mathrm{Cov}(x,y) \\ \mathrm{Cov}(x,y) & \mathrm{Var}(y) \end{bmatrix}$$

- In terms of the samples (x_i, y_i)

$$M = \frac{1}{m} \begin{bmatrix} \sum_{i=1}^{m} (x_i - \bar{x})^2 & \sum_{i=1}^{m} (x_i - \bar{x})(y_i - \bar{y}) \\ \sum_{i=1}^{m} (x_i - \bar{x})(y_i - \bar{y}) & \sum_{i=1}^{m} (y_i - \bar{y})^2 \end{bmatrix}$$

where $\bar{x} = \sum_{i=1}^{m} x_i$ and $\bar{y} = \sum_{i=1}^{m} y_i$ are the empirical means

Principal Component Analysis

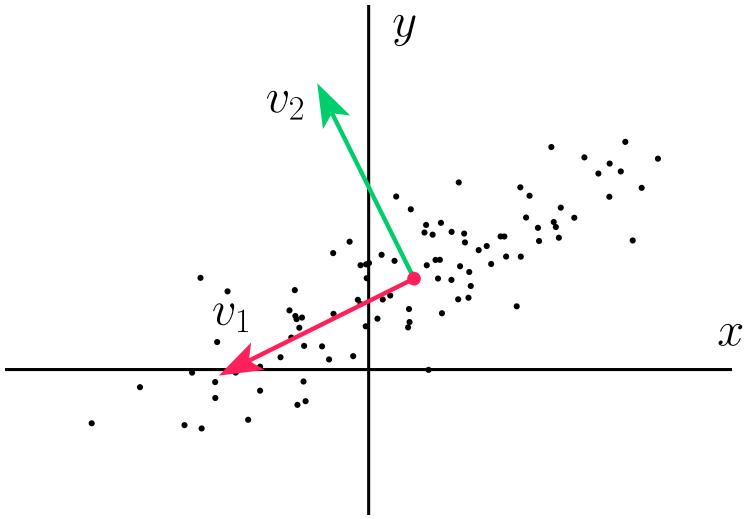
- M is a symmetric positive-definite matrix
- Variance in the direction $v \in \mathbb{R}^2$ is given by $v^T M v$
- $v^T M v$ is maximized if v is the eigenvector of M corresponding to the largest eigenvalue
- Define a matrix $A \in \mathbb{R}^{m \times 2}$

$$A = \begin{bmatrix} x_1 - \bar{x} & y_1 - \bar{y} \\ x_2 - \bar{x} & y_2 - \bar{y} \\ \vdots & \vdots \\ x_m - \bar{x} & y_m - \bar{y} \end{bmatrix}$$

- Then $M = \frac{1}{m} A^T A$

Principal Component Analysis

- From the full SVD $A = U\Sigma V^T$,
  the columns of V are the eigenvectors of $M = \frac{1}{m}A^T A$
- Define the new axes along v_1 and v_2

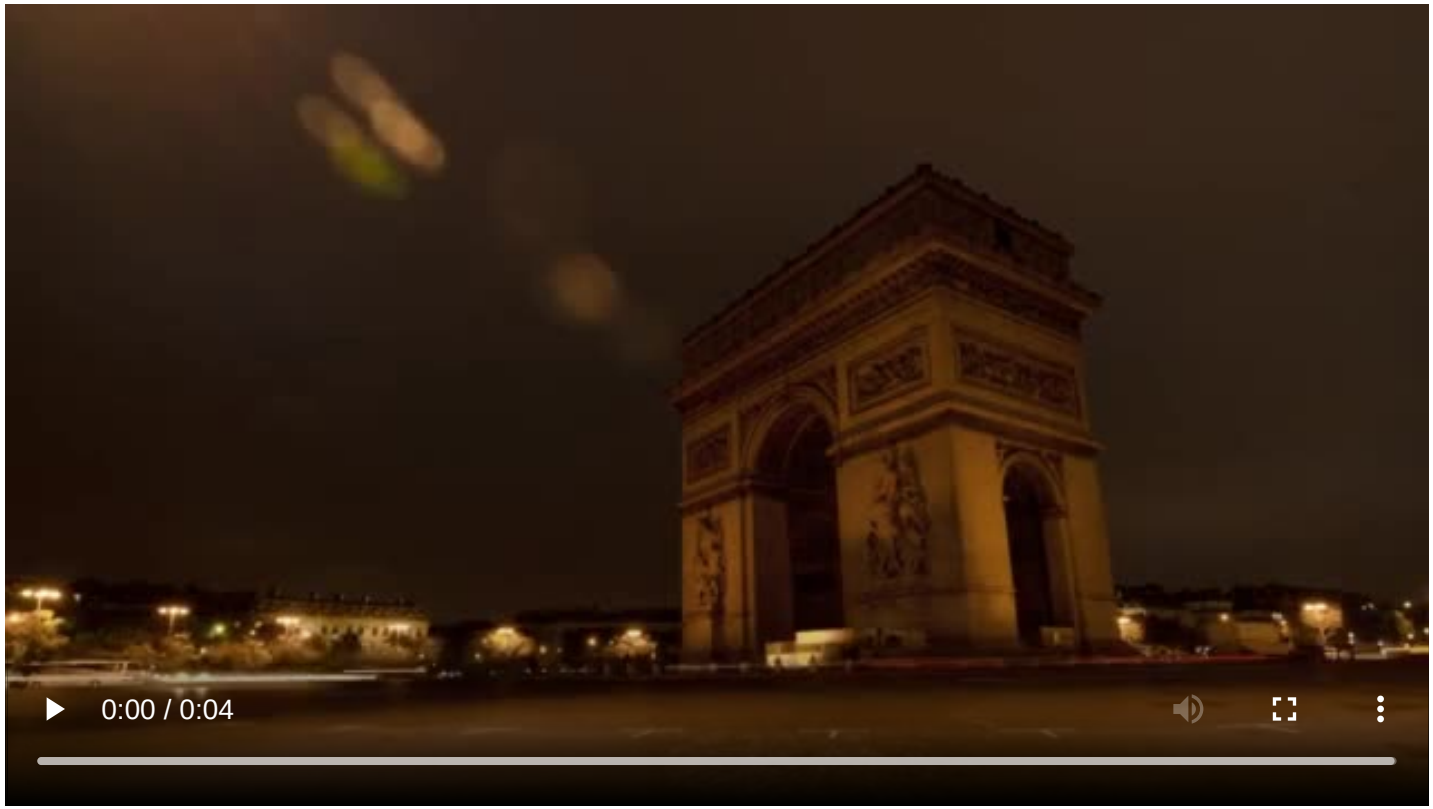


- See [examples/unit2/pca.py]

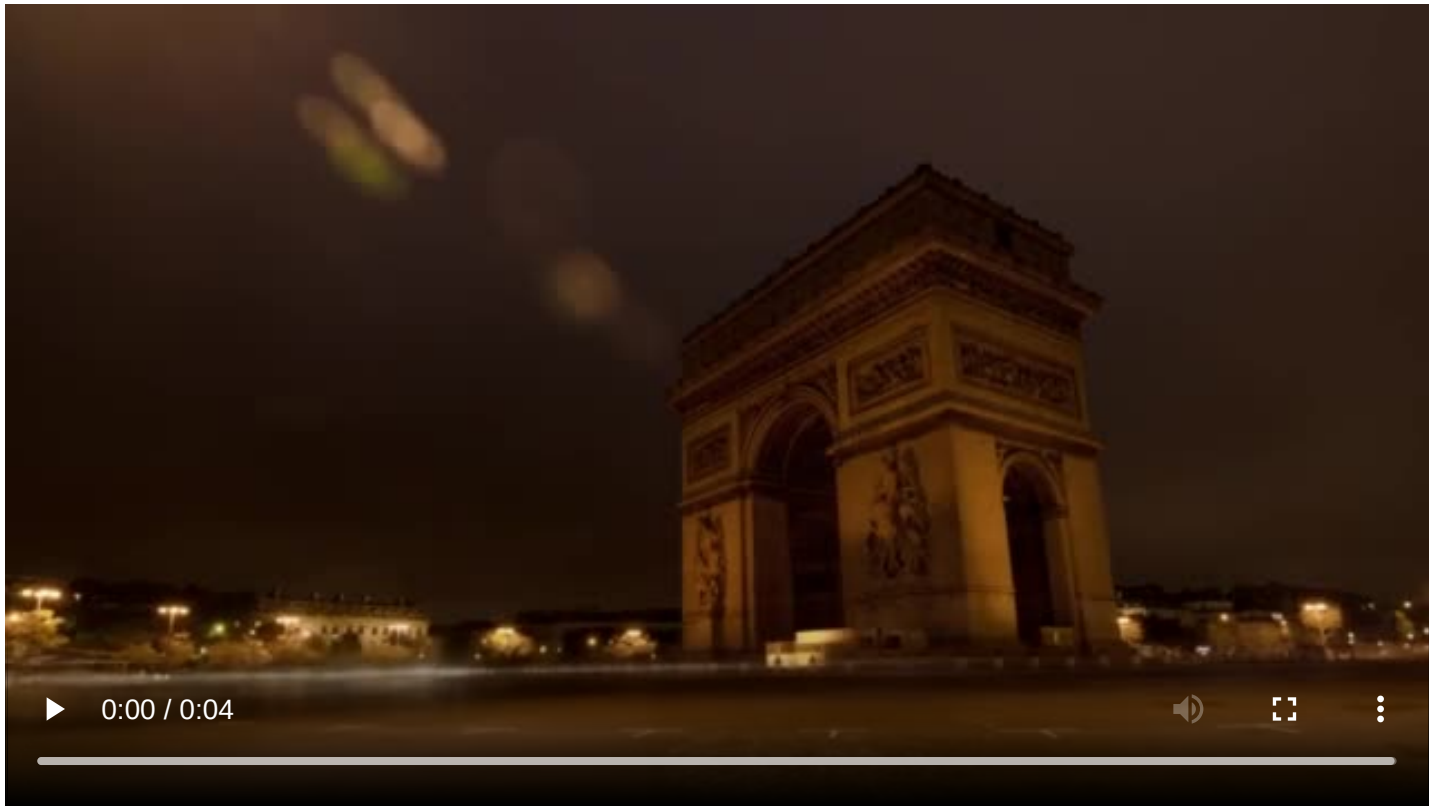
Example: Video Reconstruction

- Three videos
  - Paris https://www.pexels.com/video/852352
  - Vietnam https://www.youtube.com/watch?v=OiqSsE0B-Rc
  - Sunrise https://www.pexels.com/video/855646
- PCA applied to frames of the videos

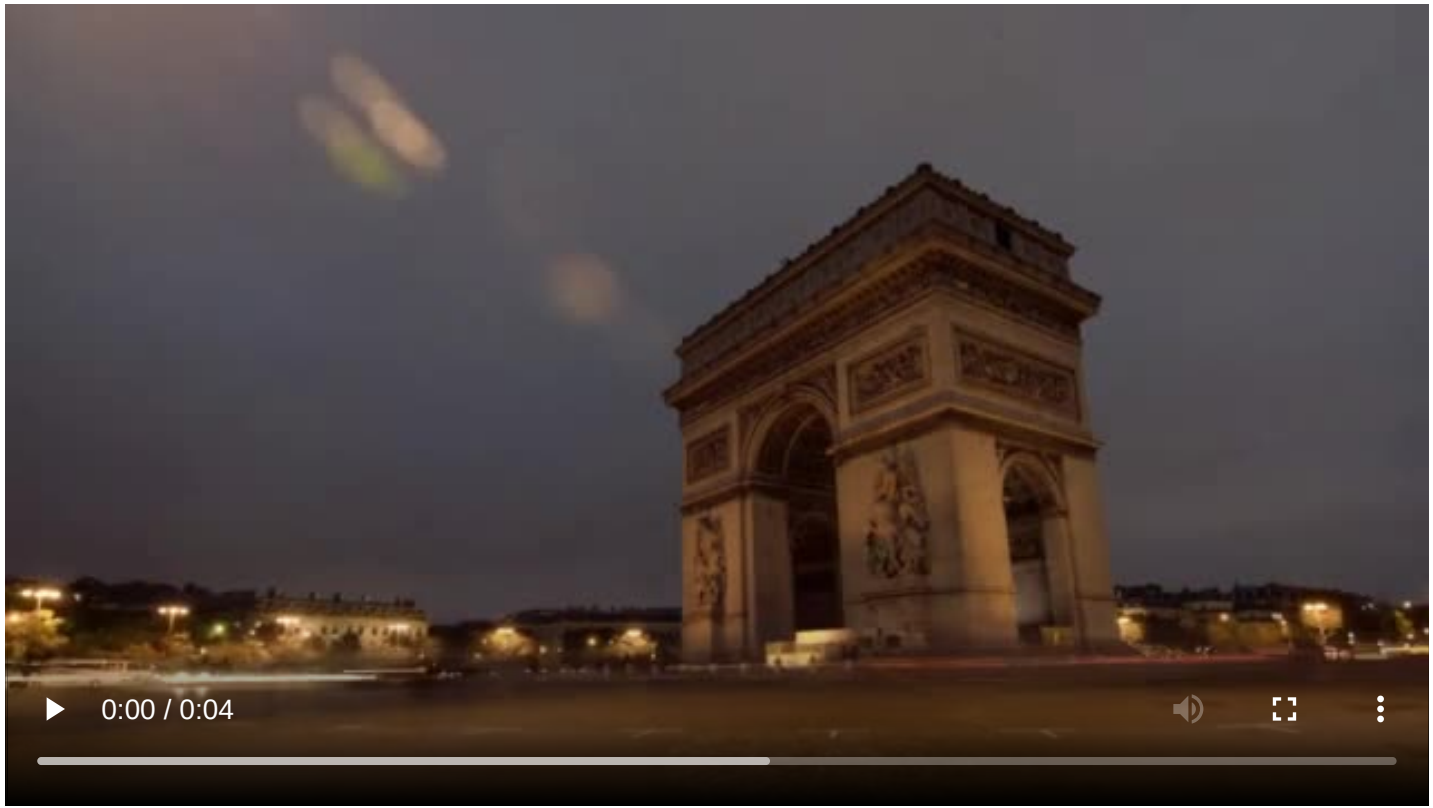
Paris, original



Paris, only first three

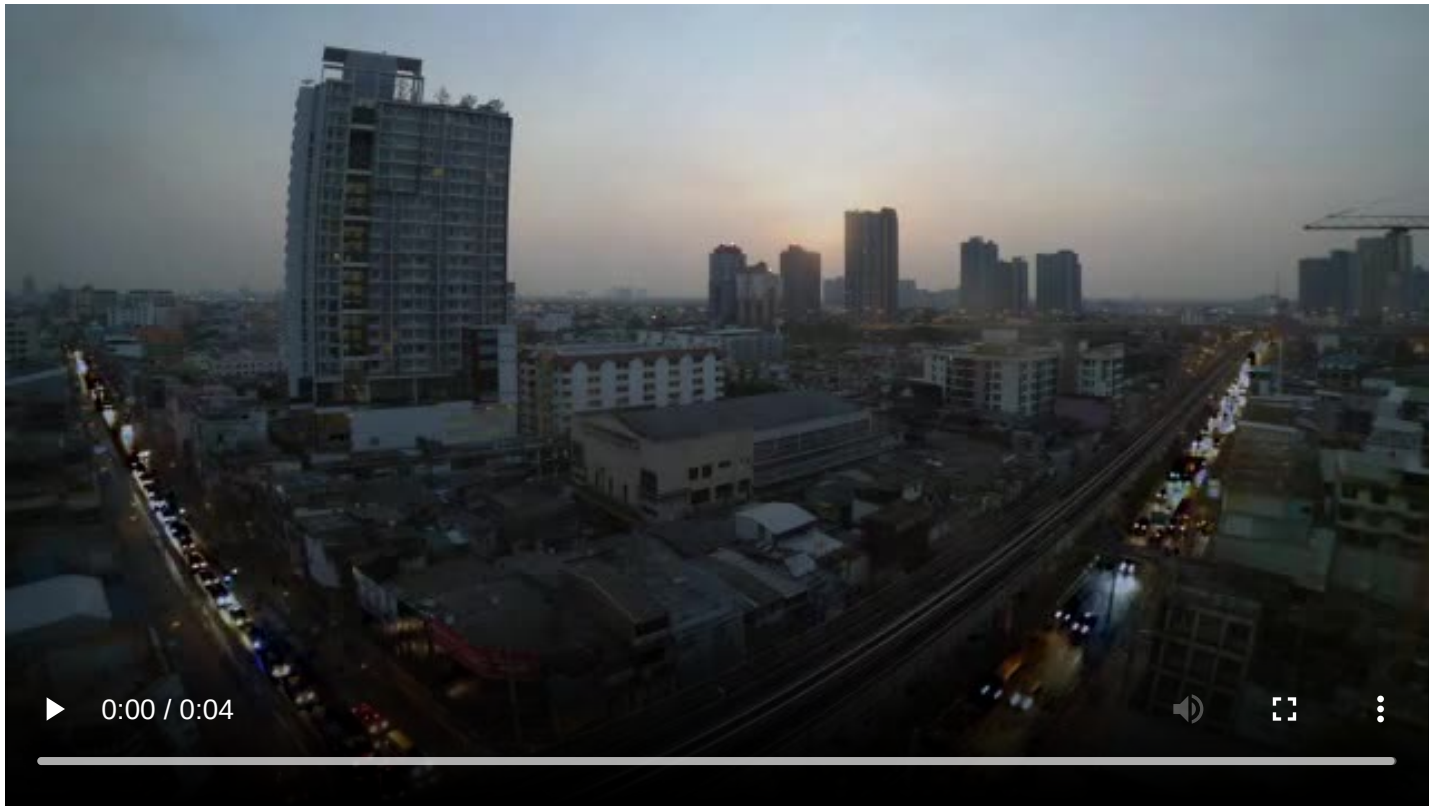


Paris, without first three



Vietnam, original

Vietnam, without first three

Sunrise, original

Sunrise, without first three